

claude-windows / 6e9d0958-a2ec-4a8e-bb2d-6cce9833c527

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\subagents\workflows\wf_8d68d336-288\agent-adb136a5d6be3ed6c.jsonl`  
- SHA-256: `307b2fcd167d2255c0aeca432bcd737b522fa82604f8dd5aca1d166759a272`  
- Source modified: `2026-07-03T19:51:15+00:00`  
- Imported at: `2026-07-05T16:48:26+00:00`  
- project: `wf_8d68d336-288`  
- session_id: `6e9d0958-a2ec-4a8e-bb2d-6cce9833c527`
```

Transcript

1. user (2026-07-03T19:48:21.200Z)

Repo (my in-progress optimization branch): C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/fix-worktree

Key files to Read:

- backend/pipeline/detectors/wasb_infer.py ? the inference loop (DataLoader ? detector.run_tensor ? tracker). Contains the streaming SequentialFrameStore, the _PrefetchIter I just added, build_windows, and the tracker re-run loop.
- backend/pipeline/detectors/native_wasb_runner.py ? builds the wasb_infer.py argv on the GPU box (native runner). Has device/stream_video/batch_size/prefetch_batches config.
- backend/config/models.py (WasbIndexConfig) and backend/config/presets.py (cloud-serving preset).
- deploy/cloudrun/Dockerfile ? the WASB conda env is torch 1.11.0+cu113 (py3.8), a SEPARATE env from the app's torch. The detector shells out to it as a subprocess.

The WASB model itself lives at /opt/models/WASB-SBDT (nttcom/WASB-SBDT) ? you can't read it, but reason from how wasb_infer.py calls build_detector(cfg) / detector.run_tensor(imgs, trans) and build_tracker(cfg).

MEASURED BASELINE (first real L4 cloud run, 2026-07-03): a ~7.5-min 60fps badminton clip took Phase-2 (WASB detection) 727.9s (~0.6x realtime); GPU utilization sat <1% the whole time. Hardware: NVIDIA L4 (Ada sm_89, 23GB), 8 vCPU, 32GB. Streaming mode forces DataLoader num_workers=0 (single-process), so CPU batch assembly (cv2 decode + PIL + ToTensor) serializes with GPU inference.

CRITICAL CONSTRAINT ? the D5/D16 byte-parity guarantee: the repo REQUIRES that the rally/window OUTPUT (the candidate windows and trajectory CSV) stay bit-identical / deterministic across runs and across the disk-frames vs streaming paths. This is asserted by a real parity test. An optimization that changes detector NUMERICS (FP16, TF32, cudnn.benchmark algo selection, NVDEC vs CPU decode) risks changing the output and breaking that guarantee. Batch size and prefetch do NOT change per-frame numerics (same forward, just grouped/overlapped) so they are parity-safe. Already implemented on this branch: indexing.wasb.batch_size + prefetch_batches knobs, the cloud-serving preset opts into batch 64 + prefetch 4.

You are a CUDA/PyTorch performance engineer. Investigate ONE optimization lever in depth by READING the actual code. Lever "tf32-cudnn":

Investigate torch.backends.cuda.matmul.allow_tf32, torch.backends.cudnn.allow_tf32, and torch.backends.cudnn.benchmark=True. TF32 accelerates conv/matmul on Ampere+/Ada with a small precision change. cudnn.benchmark autotunes algos for a fixed input size (WASB has fixed input_wh) ? big for a long fixed-size run, but the autotune picks possibly-nondeterministic algos. For EACH of the three flags: speed win, parity risk (numeric-changing vs determinism-changing), and whether it is safe-on / opt-in / unsafe given the byte-parity guarantee.

Read the named files, reason from the real code, and return the structured finding. Be concrete

about code_locations (file:function). Be honest about parity_risk ? the byte-parity guarantee is load-bearing.

2. assistant (2026-07-03T19:48:24.197Z)

I'll investigate the tf32-cudnn lever by reading the actual code to understand how WASB inference works and where these flags would apply.

3. user (2026-07-03T19:48:28.770Z)

```
1      """Linux-native WASB shuttle-trajectory runner (no WSL).
2
3      The de-risk goal the ``DetectorRunner`` ABC was built for (see ``base.py`` ?
4      "Optional Linux-native path"; ``docs/DECOUPLED_COMPUTE/04-BUILD-AND-TEST-PLAN.md``
M3.5).
5      It is the sibling of :class:`~backend.pipeline.detectors.wasb_runner.WasbRunner` for a
6      Linux GPU box (GCP / a SaaS worker): it runs the same verified ``wasb_infer.py``
core
7      that the WSL adapter runs, so the trajectory CSV is identical (to a sub-pixel float
tolerance;
8      see the acceptance gate below) ? but WITHOUT the WSL transport:
9
10     * NO ``wslCommandMixin`` (no ``wsl -d <distro> bash -c``), NO ``conda activate`` (the
11     ``wasb`` env's python is invoked by path), NO ``/mnt/c`` path translation
12     (``to_wsl_mnt_path``). Paths are native; the video is already local on the box.
13     * ``device`` is ``cuda`` (default ? the WSL parity path), ``mps``, ``cpu``, or
``auto``
14     (M4: cuda-if-available-else-cpu ? the CPU fallback for a no-GPU box; explicit
``cuda`` stays
15     strict); threaded into ``wasb_infer.py --device`` (and so the Hydra
``runner.device`` override).
16     * weights / repo / python / scratch come from the environment first
17     (``WASB_WEIGHTS`` / ``WASB_REPO_DIR`` / ``WASB_PYTHON`` / ``WASB_SCRATCH``), then
18     ``indexing.wasb.*`` config, then a default ? so a box needs no config edits.
19
20     It emits the identical ``Frame,Visibility,X,Y`` CSV (same ``{stem}__{vid12}_wasb.csv``
21     name as the WSL runner) and reuses the model-agnostic ``parse_trajectory_csv`` /
22     ``trajectory_to_action_windows`` from ``tracknet_runner``, so the trajectory hybrids and
23     the windowing brain are unchanged. Selected via the existing ``detector_impl`` seam
24     (``trajectory_hybrid._resolve_runner`` ? ``_build_native_runner``);
``detector_impl="native"``.
25
26     Acceptance gate (parity, NOT strict byte-equality): the SAME clip through the WSL
runner
27     (Windows) and this runner (Linux, ``device=cuda``) yields the same trajectory CSV + the
same
28     candidate windows within a small float tolerance (~1e-3 px). cuDNN is non-
deterministic
29     ACROSS GPUs, so a sub-pixel diff is expected on different hardware ? but it is byte-
exact
30     run-to-run on a FIXED GPU. ? Validated 2026-06-20 on an RTX 2070 (via WSL): native
streaming
31     vs the cached WSL-disk CSV matched 1194/1195 frames (the lone diff a ~5e-5 px edge
32     artifact), and two native runs were byte-identical (deterministic). This is a hardware
check
33     (a real GPU + the ``wasb`` env), so the offline, subprocess-mocked suite asserts the
contract,
34     not the numbers.
35     """
36
37     import logging
38     import os
39     import shutil
40     import subprocess
41     from dataclasses import dataclass
```

```

42     from typing import List, Optional, Tuple
43
44     from backend.config.models import IndexingConfig
45     from backend.pipeline.detectors.base import DetectorRunner
46     from backend.pipeline.detectors.device import AUTO, VALID_DEVICES, DeviceContext
47
48     # Reuse the model-agnostic helpers ? trajectory shape + windowing are identical to
WSL/TrackNet.
49     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
50
51     logger = logging.getLogger(__name__)
52
53     # Path to the inference core we copy into the WASB repo's src/ (so it can import WASB's
54     # detectors/trackers/dataloaders + find configs/), exactly as the WSL adapter does.
55     _INFER_PY = os.path.join(os.path.dirname(os.path.abspath(__file__)), "wasb_infer.py")
56
57     # Single torch/cuda probe code, used by BOTH healthcheck and the lazy device-resolution
probe
58     # so the "cuda True/False" string they parse can never drift apart.
59     _CUDA_PROBE_CODE = (
60         "import torch; print('torch', torch.__version__, 'cuda', torch.cuda.is_available())"
61     )
62
63
64     @dataclass
65     class NativeWasbConfig:
66         """Resolved settings for a Linux-native WASB run.
67
68         Built via :meth:`from_indexing_cfg` with **environment overrides** taking precedence
69         over ``indexing.wasb.*`` config (the box sets env; no config edits needed).
``device``
70         defaults to ``cuda`` (the WSL parity path); ``stream_video`` defaults to True (the
perf win).
71         """
72
73         repo_dir: str = "~/models/WASB-SBDT" # native clone of nttcom/WASB-SBDT
74         python_bin: str = (
75             "python" # the `wasb` conda env python (invoked by path, no activate)
76         )
77         weights_path: str = "" # REQUIRED (env WASB_WEIGHTS or indexing.wasb.weights_path)
78         sport: str = "badminton"
79         device: str = "cuda" # auto | cuda | mps | cpu (auto = cuda-if-available-else-cpu)
80         scratch_dir: str = "" # frame-extraction scratch (env); "" = next to the output dir
81         timeout_sec: int = 1800 # 30 min; a hung GPU call is killed (0 = no timeout)
82         keep_frames: bool = (
83             False # retain the decoded frame cache after success (debug only)
84         )
85         # NATIVE DEFAULT = STREAMING: skip the cv2 PNG extraction that dominates a long
clip's
86         # wall-clock (measured ~50s on a 20s clip) ? bit-identical to disk (verified on a
real GPU
87         # 2026-06-20). The GPU box reads from a local NVMe so streaming is the right default
here.
88         stream_video: bool = True
89         # GPU-feed tuning (indexing.wasb.batch_size / prefetch_batches). 0 = omit the flag ?
90         # wasb_infer.py's parity defaults (batch 8, no prefetch). The cloud-serving preset
opts in.
91         batch_size: int = 0
92         prefetch_batches: int = 0
93
94     @classmethod
95     def from_indexing_cfg(cls, idx_cfg: "IndexingConfig") -> "NativeWasbConfig":
96         if isinstance(idx_cfg, dict):
97             idx_cfg = IndexingConfig(**idx_cfg)
98             w = idx_cfg.wasb

```

```

99         env = os.environ.get
100         # stream_video is tri-state in config: None/absent => native default (stream);
an
101         # explicit True/False still wins (e.g. False for a container cv2 can't seek).
102         sv = w.stream_video
103         return cls(
104             repo_dir=env("WASB_REPO_DIR") or w.repo_dir,
105             python_bin=env("WASB_PYTHON") or w.python_bin,
106             weights_path=env("WASB_WEIGHTS") or w.weights_path or "",
107             sport=w.sport,
108             device=env("WASB_DEVICE") or w.device,
109             scratch_dir=env("WASB_SCRATCH")
110             or env("RALLY_SCRATCH")
111             or env("TMPDIR")
112             or "",
113             timeout_sec=w.timeout_sec,
114             keep_frames=w.keep_frames,
115             stream_video=(True if sv is None else bool(sv)),
116             batch_size=int(w.batch_size or 0),
117             prefetch_batches=int(w.prefetch_batches or 0),
118         )
119
120
121     class NativeWasbRunner(DetectorRunner):
122         """WASB runner that executes natively on a Linux GPU box ? no WSL. See module
docstring."""
123
124         def __init__(self, cfg: NativeWasbConfig):
125             self.cfg = cfg
126             # Cached CUDA-availability probe (used to resolve device="auto"). Set by
healthcheck;
127             # lazily probed by run_predict if healthcheck was skipped. None = not yet
probed.
128             self._cuda_available: Optional[bool] = None
129
130             # --- low-level native exec (the single place tests patch)
-----
131         def _run(
132             self, argv: List[str], cwd: Optional[str] = None
133         ) -> subprocess.CompletedProcess:
134             logger.debug("native exec: %s (cwd=%s)", " ".join(argv), cwd)
135             return subprocess.run(
136                 argv,
137                 cwd=cwd,
138                 capture_output=True,
139                 text=True,
140                 timeout=(self.cfg.timeout_sec or None),
141             )
142
143         # --- device resolution (M4 DeviceContext: device="auto" ? cuda-if-available-else-
cpu) -----
144         def _probe_cuda(self) -> bool:
145             """Probe ``torch.cuda.is_available()`` in the wasb env's python (a subprocess).
Any failure
146             (missing python / timeout / import error) is treated as 'no CUDA'."""
147             try:
148                 res = self._run([self.cfg.python_bin, "-c", _CUDA_PROBE_CODE])
149             except (FileNotFoundError, subprocess.TimeoutExpired):
150                 return False
151             return res.returncode == 0 and "cuda True" in (res.stdout or "")
152
153         def _cuda_is_available(self) -> bool:
154             """CUDA availability, cached across healthcheck + run_predict so we probe at
most once."""
155             if self._cuda_available is None:

```

```

156         self._cuda_available = self._probe_cuda()
157         return self._cuda_available
158
159     def _effective_device(self) -> str:
160         """The torch device to actually pass to ``wasb_infer.py``. Only ``auto`` needs
the CUDA
161         probe; an explicit cuda/mps/cpu resolves to itself (so ``device='cuda'`` is
unchanged)."""
162         if self.cfg.device == AUTO:
163             return DeviceContext(AUTO, self._cuda_is_available()).effective
164         return self.cfg.device
165
166     def _repo_src(self) -> str:
167         return os.path.join(os.path.expanduser(self.cfg.repo_dir), "src")
168
169     def _copy_infer_script(self) -> bool:
170         """Copy wasb_infer.py into the WASB repo src/ so it imports WASB modules + finds
configs/."""
171         repo_src = self._repo_src()
172         try:
173             os.makedirs(repo_src, exist_ok=True)
174             shutil.copy(_INFER_PY, os.path.join(repo_src, "wasb_infer.py"))
175             return True
176         except OSError as e:
177             logger.error("Failed to copy wasb_infer.py into %s: %s", repo_src, e)
178             return False
179
180     def _rm_rf(self, path: Optional[str]) -> None:
181         """Best-effort recursive delete of a native path (post-success cleanup; never
raises)."""
182         if path:
183             shutil.rmtree(path, ignore_errors=True)
184
185     def healthcheck(self) -> Tuple[bool, str]:
186         """Verify weights + repo present and the `wasb` python imports torch (and sees a
GPU on cuda)."""
187         if not self.cfg.weights_path:
188             return False, (
189                 "WASB weights path is empty ? set WASB_WEIGHTS (or
indexing.wasb.weights_path).",
190             )
191         weights = os.path.expanduser(self.cfg.weights_path)
192         if not os.path.exists(weights):
193             return False, f"WASB weights not found at {weights}"
194         repo_src = self._repo_src()
195         if not os.path.isdir(repo_src):
196             return False, (
197                 f"WASB-SBDT repo src/ not found at {repo_src} ? clone nttcom/WASB-SBDT "
198                 f"(set WASB_REPO_DIR / indexing.wasb.repo_dir).",
199             )
200         # Fail fast on a typo'd device (the device-policy seam) ? a clear error here
beats a cryptic
201         # argparse rc=2 from wasb_infer.py at run time.
202         if self.cfg.device not in VALID_DEVICES:
203             return False, (
204                 f"unknown device {self.cfg.device!r} ? valid: {'',
'.join(VALID_DEVICES)}.",
205             )
206         try:
207             res = self._run([self.cfg.python_bin, "-c", _CUDA_PROBE_CODE])
208         except FileNotFoundError:
209             return (
210                 False,
211                 f"python binary not found: {self.cfg.python_bin!r} (set WASB_PYTHON).",
212             )

```

```

213         except subprocess.TimeoutExpired:
214             return False, "native torch healthcheck timed out."
215         out = (res.stdout or "").strip()
216         if res.returncode != 0:
217             return (
218                 False,
219                 f"{self.cfg.python_bin}` torch import failed: {(res.stderr or
out).strip()}",
220             )
221         # M4: resolve the device. Cache the probe so run_predict reuses it (no second
subprocess).
222         self._cuda_available = "cuda True" in out
223         ctx = DeviceContext(self.cfg.device, self._cuda_available)
224         if ctx.cuda_required_but_missing:
225             # Unchanged pre-M4 behaviour for an EXPLICIT cuda request: hard-fail, never
a silent
226             # slow-CPU downgrade. Set indexing.wasb.device=auto for a CPU fallback.
227             return False, f"device=cuda but torch.cuda.is_available() is False ({out})"
228         if ctx.fell_back:
229             logger.warning(
230                 "device=auto: no CUDA GPU detected (%s) ? FALLING BACK TO CPU; WASB "
231                 "inference will be slow. Set indexing.wasb.device=cuda to require a
GPU.",
232                 out,
233             )
234         return True, out
235
236     def run_predict(
237         self, video_win_path: str, output_win_dir: str, video_id: Optional[str] = None
238     ) -> Optional[str]:
239         """Run WASB natively on a video. Returns the path to the trajectory CSV, or
None.
240
241         Output artifacts are namespaced by ``video_id`` (the file MD5) exactly as the
WSL
242         runner does, and the CSV is named ``{stem}__{vid12}_wasb.csv`` ? byte-for-byte
the
243         same downstream contract, so the parity comparison is apples-to-apples.
244         """
245         output_dir = os.path.abspath(output_win_dir)
246         os.makedirs(output_dir, exist_ok=True)
247         # Normalize for a cross-platform basename (tests pass Windows paths on Linux).
248         norm = str(video_win_path).replace("\\", "/")
249         stem, _ext = os.path.splitext(os.path.basename(norm))
250         # Content-derived key: same filename + different bytes -> different key (no
cache reuse).
251         key = f"{stem}__{video_id[:12]}" if video_id else (stem or "wasb")
252         out_csv = os.path.join(output_dir, f"{key}_wasb.csv")
253
254         if not self._copy_infer_script():
255             logger.error("Could not stage wasb_infer.py into the WASB repo src/.")
256             return None
257
258         weights = os.path.expanduser(self.cfg.weights_path)
259         argv = [
260             self.cfg.python_bin,
261             "wasb_infer.py",
262             "--video",
263             os.path.abspath(str(video_win_path)),
264         ]
265         frames_dir: Optional[str] = None
266         if self.cfg.stream_video:
267             # Fast path: decode in-process, no PNG extraction. Output is bit-identical
to disk.
268             argv.append("--stream-video")

```

```

269         else:
270             scratch = self.cfg.scratch_dir or output_dir
271             frames_dir = os.path.join(scratch, f"{key}_frames")
272             argv += ["--frames_out_dir", frames_dir]
273         # M4: resolve device="auto" to the effective device (cuda-if-available-else-
cpu); an
274         # explicit cuda/mps/cpu passes through unchanged. Never sends "auto" to
wasb_infer.py.
275         device = self._effective_device()
276         argv += [
277             "--weights",
278             weights,
279             "--sport",
280             self.cfg.sport,
281             "--device",
282             device,
283             "--out",
284             out_csv,
285         ]
286         # GPU-feed tuning: pass only explicit non-zero values so the default argv (and
its
287         # byte-parity guarantee) is unchanged when the knobs are unset.
288         if self.cfg.batch_size:
289             argv += ["--batch-size", str(self.cfg.batch_size)]
290         if self.cfg.prefetch_batches:
291             argv += ["--prefetch-batches", str(self.cfg.prefetch_batches)]
292
293         logger.info(
294             "Running native WASB inference (device=%s%s) on %s ...",
295             device,
296             " [auto?fallback]" if self.cfg.device == AUTO and device != "cuda" else "",
297             os.path.basename(norm),
298         )
299         try:
300             res = self._run(argv, cwd=self._repo_src())
301         except FileNotFoundError:
302             logger.error(
303                 "python binary not found: %r (set WASB_PYTHON).", self.cfg.python_bin
304             )
305             return None
306         except subprocess.TimeoutExpired:
307             logger.error(
308                 "WASB native inference timed out after %ss.", self.cfg.timeout_sec
309             )
310             return None
311         if res.returncode != 0:
312             logger.error(
313                 "WASB native inference failed:\n%s",
314                 (res.stderr or res.stdout or "")[-2000:],
315             )
316             return None
317         if not os.path.exists(out_csv):
318             logger.error(
319                 "WASB native finished but expected CSV not found at %s", out_csv
320             )
321             return None
322         logger.info("WASB trajectory CSV: %s", out_csv)
323
324         # Success ? the CSV is the durable output; the decoded frame cache is a pure
intermediate.
325         # On any earlier failure we return above WITHOUT cleaning, so the cache survives
a resume.
326         # (Streaming never materializes frames, so there is nothing to drop on that
path.)
327         if not self.cfg.keep_frames and frames_dir:

```

```

328         logger.info(
329             "Cache hygiene: removing frame cache for %s "
330             "(set indexing.wasb.keep_frames=true to retain).",
331             key,
332         )
333         self._rm_rf(frames_dir)
334     return out_csv
335
336     # Trajectory parsing + windowing are inherited from tracknet_runner (model-
agnostic).
337     parse_trajectory_csv = staticmethod(TrackNetRunner.parse_trajectory_csv)
338     trajectory_to_action_windows = staticmethod(
339         TrackNetRunner.trajectory_to_action_windows
340     )
341

```

4. user (2026-07-03T19:48:29.014Z)

```

1     """WASB shuttle-trajectory inference on an arbitrary frame sequence (runs in WSL).
2
3     WASB-SBDT only ships *dataset* evaluation; this is a thin wrapper that reuses its
4     own building blocks (ImageDataset transform + detector + online tracker) to run on
5     any folder of extracted frames and emit a trajectory CSV in the exact shape the
6     indexer's `tracknet_runner.parse_trajectory_csv` already consumes:
7
8         Frame,Visibility,X,Y
9
10    It must run INSIDE the WSL `wasb` conda env, from the WASB-SBDT `src/` dir (it
11    imports WASB's `detectors`/`trackers`/`dataloaders`). The Windows-side adapter
12    (`wasb_runner.py`) stages this file + the frames into WSL and invokes it.
13
14    Resumable across reboots
15    -----
16    The GPU detector pass (one forward per sliding window ? ~21k for a 6-min clip) is
17    the slow, expensive part; the CPU tracker pass that turns detections into a
18    trajectory is cheap and *stateful* (it must see every frame in order, so it can't
19    resume mid-stream). We exploit that split:
20
21    * The detector pass writes its per-window raw outputs incrementally to
22      ``<cache>/det_raw.jsonl`` (flushed+fsync'd every ``--flush-every`` windows) and
23      records progress in ``<cache>/manifest.json`` (atomic write). A reboot loses at
24      most ``--flush-every`` windows of GPU work instead of the whole pass.
25    * On restart we replay the cached windows, skip the DataLoader ahead to the first
26      un-cached window, and continue. Once every window is cached, the detector pass
27      is skipped entirely and we just re-run the (cheap, deterministic) tracker over
28      the full ordered cache -> trajectory CSV. So a lost trajectory CSV costs seconds,
29      not the GPU hour.
30
31    The cache is keyed by the output path (``<out>_wasbcache/`` by default), lives next
32    to the output (NOT in %TEMP%), and is invalidated automatically if the frames dir,
33    weights, sport, window size, or frame count change.
34
35    Usage (in WSL, from ~/models/WASB-SBDT/src):
36    python wasb_infer.py --frames_dir /path/frames --weights
../pretrained_weights/wasb_badminton_best.pth.tar \
37    --out /path/traj.csv [--sport badminton] [--limit N] [--cache-dir DIR] [--flush-
every 1000] [--fresh]
38    """
39
40    import argparse
41    import csv
42    import glob
43    import json
44    import logging
45    import os

```



```

46     import os.path as osp
47     from collections import defaultdict
48     from contextlib import contextmanager
49
50     logger = logging.getLogger(__name__)
51
52     CACHE_VERSION = 1
53     _DET_FILE = "det_raw.jsonl"
54     _MANIFEST_FILE = "manifest.json"
55
56
57     def _build_cfg(weights: str, sport: str, device: str = "cuda"):
58         """Compose the WASB eval config via Hydra, overriding model/weights/device.
59
60         ``device`` is ``cuda`` (default ? the historical WSL behaviour, byte-parity
61         preserved), ``mps`` (Apple) or ``cpu``. Only the single-GPU CUDA path requests
62         ``runner.gpus=[0]``;
63         """
64         from hydra import compose, initialize
65
66         gpus = "[0]" if device == "cuda" else "[]" # single GPU on cuda; none for cpu/mps
67         with initialize(config_path="configs", version_base=None):
68             cfg = compose(
69                 config_name="eval",
70                 overrides=[
71                     f"dataset={sport}",
72                     "model=wasb",
73                     f"detector.model_path={weights}",
74                     f"runner.device={device}",
75                     f"runner.gpus={gpus}", # default config assumes multiple GPUs; pin to
76 this box
77             ],
78             )
79         return cfg
80
81     def _frame_id(path: str) -> int:
82         """Best-effort integer frame id from a frame filename (e.g. frame_000180.png ->
83 180)."""
84         stem = osp.splitext(osp.basename(path))[0]
85         digits = "".join(ch for ch in stem if ch.isdigit())
86         return int(digits) if digits else -1
87
88     # ----- #
89     # Resumable detector cache (pure-Python, no torch/numpy ? unit-testable)
90     # ----- #
91     def _cache_paths(cache_dir: str):
92         return osp.join(cache_dir, _DET_FILE), osp.join(cache_dir, _MANIFEST_FILE)
93
94
95     def default_cache_dir(out_csv: str) -> str:
96         """Stable cache dir next to the trajectory output (survives reboots; not %TEMP%)."""
97         d = osp.dirname(osp.abspath(out_csv))
98         stem = osp.splitext(osp.basename(out_csv))[0]
99         return osp.join(d, stem + "_wasbcache")
100
101
102     def _atomic_write_text(path: str, text: str) -> None:
103         """Write then rename so a crash mid-write can't leave a half-written file."""
104         tmp = path + ".tmp"
105         with open(tmp, "w") as f:
106             f.write(text)

```

```

107         f.flush()
108         os.fsync(f.fileno())
109         os.replace(tmp, path)
110
111
112     def _det_plain(det) -> list:
113         """A detector candidate dict {'xy': np.array([x,y]), 'score', 'scale'} -> JSON-able
114         list."""
115         xy = det["xy"]
116         return [float(xy[0]), float(xy[1]), float(det["score"]), int(det["scale"])]
117
118     def _dets_to_tracker(plain_list):
119         """Inverse of _det_plain: rebuild the dicts the tracker expects (xy MUST be a numpy
120         array,
121         the tracker does vector math / np.linalg.norm on it)."""
122         import numpy as np
123
124         return [
125             {"xy": np.array([p[0], p[1]], dtype=float), "score": p[2], "scale": p[3]}
126             for p in plain_list
127         ]
128
129     def write_manifest(cache_dir: str, manifest: dict) -> None:
130         _, mpath = _cache_paths(cache_dir)
131         _atomic_write_text(mpath, json.dumps(manifest, indent=2))
132
133
134     def read_manifest(cache_dir: str):
135         _, mpath = _cache_paths(cache_dir)
136         if not osp.exists(mpath):
137             return None
138         try:
139             with open(mpath) as f:
140                 return json.load(f)
141         except (json.JSONDecodeError, OSError):
142             return None
143
144
145     def manifest_compatible(
146         manifest: dict,
147         *,
148         frames_dir: str,
149         weights: str,
150         sport: str,
151         frames_in: int,
152         frames_total: int,
153         device: str = "cuda",
154     ) -> bool:
155         """A cache is reusable only if the run that produced it matches this run's inputs.
156
157         ``device`` defaults to ``cuda`` so caches written before device-keying (no
158         ``device``
159         field) stay compatible with a CUDA run ? but a cpu/mps run will NOT reuse a cuda
160         cache
161         (detections can differ numerically across devices), and vice-versa.
162
163         ``frames_dir`` is the **already-canonical cache key** the caller stored in the
164         manifest
165         (``run()`` passes its ``cache_key`` here, and writes the same value into the
166         manifest):
167         ``osp.abspath(frames_dir)`` for the frames-on-disk path, or the synthetic
168         ``video:<abspath>``
169         key for the streaming path. We therefore compare it VERBATIM ? do NOT re-``abspath``

```

```

it here:
165     ``osp.abspath("video:/x.mp4")`` is not recognised as absolute and gets the CWD
prepended,
166     so a streaming cache would never match itself and every re-run would recompute
(issue #348).
167     """
168     if not manifest or manifest.get("version") != CACHE_VERSION:
169         return False
170     return (
171         manifest.get("frames_dir") == frames_dir
172         and manifest.get("weights") == weights
173         and manifest.get("sport") == sport
174         and manifest.get("frames_in") == frames_in
175         and manifest.get("frames_total") == frames_total
176         and manifest.get("device", "cuda") == device
177     )
178
179
180 def reset_cache(cache_dir: str) -> None:
181     """Drop any existing cache so the next run starts clean."""
182     det_jsonl, mpath = _cache_paths(cache_dir)
183     for p in (det_jsonl, mpath):
184         if osp.exists(p):
185             os.remove(p)
186
187
188 def load_and_clean_cache(cache_dir: str):
189     """Read the longest *contiguous* (w == line index) valid prefix of det_raw.jsonl,
190     rebuild the per-frame detection accumulation, and rewrite the file to exactly that
191     prefix so a trailing half-written line from a crash can't corrupt future appends.
192
193     Returns (windows_done, det_by_fid) where det_by_fid maps frame_id -> list of
194     [x, y, score, scale] candidates (accumulated across every window the frame is in,
195     in window order ? identical to a non-resumed run).
196     """
197     det_jsonl, _ = _cache_paths(cache_dir)
198     det_by_fid = defaultdict(list)
199     valid: list = []
200     if osp.exists(det_jsonl):
201         with open(det_jsonl, "r") as f:
202             for line in f:
203                 s = line.strip()
204                 if not s:
205                     continue
206                 try:
207                     obj = json.loads(s)
208                 except json.JSONDecodeError:
209                     break # truncated trailing line (crash mid-flush)
210                 if obj.get("w") != len(valid): # non-contiguous -> stop at the gap
211                     break
212                 valid.append(s)
213                 for fid, plain in obj["f"]:
214                     det_by_fid[int(fid)].extend([list(p) for p in plain])
215                 # Guarantee a clean append boundary by rewriting only the good prefix.
216                 _atomic_write_text(det_jsonl, ("\n".join(valid) + "\n") if valid else "")
217     return len(valid), det_by_fid
218
219
220 def _append_windows(fh, objs) -> None:
221     """Append window records as JSONL and durably flush (bounded loss on crash)."""
222     for o in objs:
223         fh.write(json.dumps(o, separators=(",", ":")) + "\n")
224     fh.flush()
225     os.fsync(fh.fileno())
226

```

```

227
228 def _atomic_write_trajectory(out_csv: str, rows) -> None:
229     os.makedirs(osp.dirname(osp.abspath(out_csv)), exist_ok=True)
230     tmp = out_csv + ".tmp"
231     with open(tmp, "w", newline="") as f:
232         w = csv.writer(f)
233         w.writerow(["Frame", "Visibility", "X", "Y"])
234         w.writerows(rows)
235         f.flush()
236         os.fsync(f.fileno())
237     os.replace(tmp, out_csv)
238
239
240 # ----- #
241 # Frame addressing + windowing (pure; no torch/cv2 ? unit-testable)
242 # ----- #
243 _STREAM_FRAME_FMT = "{:08d}.png"
244
245
246 def synthetic_frame_path(index: int) -> str:
247     """Stable, index-encoding frame name used by the streaming path.
248
249     It mirrors the on-disk names ``extract_frames`` writes (``{i:08d}.png``) so the
250     downstream integer-id parser (``_frame_id``) yields the SAME frame id whether the
251     frames came from disk or from the in-memory stream. The streaming image loader
252     inverts this name back to an index to fetch the decoded frame.
253     """
254     return _STREAM_FRAME_FMT.format(int(index))
255
256
257 def build_windows(frames: list, frames_in: int):
258     """Sliding windows of ``frames_in`` consecutive frame *paths* (the unit of GPU work
259     + checkpoint). Pure list math, identical for the disk and streaming paths.
260
261     Returns a list of windows, each a list of ``frames_in`` consecutive paths
262     (``[frames[i:i+frames_in] for i in range(len(frames) - frames_in + 1)]``). The
263     caller wraps each window into the ``{"frames", "annos"}`` sample shape WASB's
264     ``ImageDataset`` expects; keeping that out of here stays torch-free for unit tests.
265     """
266     return [frames[i : i + frames_in] for i in range(len(frames) - frames_in + 1)]
267
268
269 class SequentialFrameStore:
270     """In-memory sliding buffer over a forward-only frame reader, sized for the
271     detector's sliding-window access pattern (peak 0(window) frames, not 0(video)).
272
273     The detector DataLoader runs with ``shuffle=False`` and (in streaming mode)
274     ``num_workers=0``. ``ImageDataset.__getitem__(i)`` reads the frames of window ``i``
275     in order ? ``i, i+1, ..., i+window-1`` ? and samples are requested ``i = start,
276     start+1, ...``. So the global read sequence dips back by ``window-1`` at each window
277     boundary (``0,1,2, 1,2,3, 2,3,4, ...`` for ``window=3``); the highest index seen so
278     far never decreases. We keep only frames at or above ``max_seen - (window-1)`` (the
279     earliest index any not-yet-finished window can still ask for) and decode each source
280     frame exactly once via the forward-only ``reader(index)``.
281     """
282
283     def __init__(self, reader, total: int, window: int = 1, start_index: int = 0):
284         self._reader = reader
285         self._total = int(total)
286         self._window = max(1, int(window))
287         self._buf: dict = {}
288         # On a resumed run the detector's first needed frame is `start_index`; begin
289         # pulling there so the reader can skip (seek past) the already-cached prefix
290         # instead of decoding 0..start_index-1 just to throw them away.
291         self._next = int(start_index) # next index not yet pulled from the reader

```

```

292         self._max_seen = int(start_index) - 1
293         self._floor = int(start_index) # lowest index we still keep (never evict below)
294
295     def get(self, index: int):
296         index = int(index)
297         if index < self._floor:
298             raise KeyError(
299                 f"frame {index} already evicted (below floor {self._floor}); the access
300 "
301                 f"pattern must stay within a window of the highest frame seen"
302             )
303         if index >= self._total:
304             raise KeyError(f"frame {index} out of range (total {self._total})")
305         # Pull forward from the reader until the requested index is buffered.
306         while self._next <= index:
307             self._buf[self._next] = self._reader(self._next)
308             self._next += 1
309         if index > self._max_seen:
310             self._max_seen = index
311         # Evict frames older than the earliest a still-running window could re-request:
312         # once we've seen index `m`, no future window starts before `m - (window-1)`.
313         new_floor = max(self._floor, self._max_seen - (self._window - 1))
314         for k in range(self._floor, new_floor):
315             self._buf.pop(k, None)
316         self._floor = new_floor
317         return self._buf[index]
318
319     def _index_from_synthetic_path(path: str) -> int:
320         """Invert ``synthetic_frame_path``: a frame path -> its integer source index.
321
322         Reuses ``_frame_id`` (digits-only parse) so the index and the cached frame-id stay
323         in lockstep with the on-disk naming scheme.
324         """
325         return _frame_id(path)
326
327     def _bgr_to_pil_rgb(frame_bgr):
328         """Convert an in-memory BGR uint8 frame (as ``cv2.VideoCapture`` yields) to the
329         EXACT
330         PIL RGB image WASB's disk path produces.
331
332         The disk path is ``frame_bgr -> cv2.imwrite(PNG) ->
333         Image.open(PNG).convert('RGB')``;
334         because PNG is lossless that round-trip equals ``cv2.cvtColor(frame_bgr,
335         COLOR_BGR2RGB)``
336         bit-for-bit (verified empirically, max|diff|=0). Returning that here makes the
337         streamed
338         frame indistinguishable from the on-disk one to everything downstream.
339         """
340         import cv2
341         from PIL import Image
342
343         return Image.fromarray(cv2.cvtColor(frame_bgr, cv2.COLOR_BGR2RGB))
344
345     def _make_streamed_read_image(store, real_read_image):
346         """Build the ``read_image`` replacement used during a streaming detector pass.
347
348         A synthetic frame path -> the in-memory decoded frame for its index (as a PIL RGB
349         image,
350         matching ``read_image``'s real return type); any path that doesn't parse to an index
351         falls
352         through to ``real_read_image``. Pure ? imports no WASB module ? so it is unit-
353         testable

```

```

349 without the WSL-only ``dataloaders`` package.
350 """
351
352 def _read_image(path, *args, **kwargs):
353     idx = _index_from_synthetic_path(path)
354     if idx < 0:
355         return real_read_image(path, *args, **kwargs)
356     return _bgr_to_pil_rgb(store.get(idx))
357
358 return _read_image
359
360
361 @contextmanager
362 def _streaming_read_image_patch(
363     frame_loader, total: int, window: int = 1, start_index: int = 0
364 ):
365     """Scope a shim over WASB's ``read_image`` so ``ImageDataset`` is fed in-memory
366 decoded frames during the detector pass, byte-for-byte as it would over real PNGs.
367
368 WASB's ``ImageDataset.__getitem__`` reads each frame via ``read_image(path)`` ? NOT
369 ``cv2.imread``. ``read_image`` (``utils.utils``) does
370 ``Image.open(path).convert('RGB')``,
371 returning a PIL **RGB** image, after an ``osp.exists(path)`` guard. We therefore
372 feed it a PIL RGB image built from the streamed BGR frame (see ``_bgr_to_pil_rgb``), which is
373 byte-identical to the disk round-trip, and the shim never touches the filesystem so
374 the ``osp.exists`` guard is sidestepped for synthetic stream paths.
375
376 We patch ``dataloaders.dataset_loader.read_image`` ? the binding the consumer
377 actually calls (``dataset_loader`` does ``from utils import read_image``, so the name lives
378 in that module's namespace; patching ``utils.read_image`` would NOT rebind the already-
379 imported reference).
380
381 ``frame_loader(index) -> BGR uint8 ndarray`` is a forward-only sequential decoder
382 wrapped in a ``SequentialFrameStore`` (sliding buffer sized to ``window`` = ``frames_in``);
383 on a resume we start at ``start_index`` so the decoder seeks past already-cached windows.
384 The original reader is restored on exit.
385 """
386 from dataloaders import dataset_loader as _dl
387
388 store = SequentialFrameStore(
389     frame_loader, total, window=window, start_index=start_index
390 )
391 real_read_image = _dl.read_image
392 # Rebind read_image in the consuming module for the duration of the detector pass;
393 # restore on exit. (Plain assignment, not setattr-with-constant, to stay ruff
394 B010-clean.)
395 _dl.read_image = _make_streamed_read_image(store, real_read_image) # type:
396 ignore[assignment]
397 try:
398     yield store
399 finally:
400     _dl.read_image = real_read_image # type: ignore[assignment]
401
402 # ----- #
403 # Inference

```

```

402 # ----- #
403 class _PrefetchIter:
404     """Bounded producer-thread wrapper over a batch iterator (the num_workers=0
DataLoader).
405
406     In streaming mode the DataLoader must stay single-process (the in-memory frame store
+
407     the cv2.imread shim live in this process and cannot be pickled to workers), so the
408     CPU-side batch assembly (decode + PIL + ToTensor) SERIALIZES with GPU inference ?
409     measured on the first real L4 serving run (2026-07-03) this left the GPU <1%
utilized.
410
411     A single producer THREAD fixes the overlap without multiprocessing: cv2/PIL/numpy
412     release the GIL for the heavy parts, the store's forward-only access pattern is
413     preserved (one consumer thread iterating sequentially), and the bounded queue caps
414     memory at ``depth`` batches. A producer exception is re-raised in the consumer."""
415
416     _END = object()
417
418     def __init__(self, src, depth: int = 2):
419         import queue
420         import threading
421
422         self._q: "queue.Queue" = queue.Queue(maxsize=max(1, int(depth)))
423         self._exc: "BaseException | None" = None
424
425         def _fill() -> None:
426             try:
427                 for item in src:
428                     self._q.put(item)
429             except BaseException as e: # noqa: BLE001 - surfaced in __next__
430                 self._exc = e
431             finally:
432                 self._q.put(self._END)
433
434         self._t = threading.Thread(target=_fill, name="wasb-prefetch", daemon=True)
435         self._t.start()
436
437     def __iter__(self):
438         return self
439
440     def __next__(self):
441         item = self._q.get()
442         if item is self._END:
443             if self._exc is not None:
444                 raise self._exc
445             raise StopIteration
446         return item
447
448     def run(
449         frames,
450         weights: str,
451         out_csv: str,
452         sport: str = "badminton",
453         limit: int = 0,
454         batch_size: int = 8,
455         num_workers: int = 4,
456         log_every_batches: int = 50,
457         cache_dir: "str | None" = None,
458         flush_every: int = 1000,
459         fresh: bool = False,
460         frame_loader=None,
461         source_key: "str | None" = None,
462         device: str = "cuda",
463         prefetch_batches: int = 0,

```

```

464     ) -> str:
465         """Run WASB detection+tracking over an ordered frame source -> trajectory CSV.
466
467         Two equivalent ways to supply pixels (output is bit-identical between them):
468
469         * **frames-on-disk** (default): ``frames`` is a directory path string; frames are
470           globbed (``*.png``/``*.jpg``) and ``ImageDataset`` reads each file via
471           ``cv2.imread``.
472         * **streaming** (fast path): ``frames`` is a pre-built ordered list of (synthetic)
473           frame paths and ``frame_loader(index) -> BGR uint8 ndarray`` supplies the decoded
474           pixels in-memory. We scope a shim over WASB's ``read_image`` (the actual per-frame
475           reader, which returns a PIL RGB image) over the DataLoader pass so every other
476           line of
477           ``ImageDataset.__getitem__`` runs UNCHANGED ? the tensor the detector sees is
478           identical
479           to the disk round-trip (``cv2.imwrite`` PNG -> ``Image.open().convert('RGB')`` ==
480           ``cvtColor(bgr, BGR2RGB)``), verified byte-identical). Streaming forces
481           ``num_workers=0``
482           (the in-process shim + buffer can't cross worker processes).
483
484           ``source_key`` overrides the cache-keying identity (defaults to the abspath of the
485           frames dir); the streaming path passes a stable ``video:<abspath>`` key so a resume
486           after reboot still matches.
487         """
488         import time
489
490         import torch
491         from dataloaders import build_img_transforms, build_seq_transforms
492         from dataloaders.dataset_loader import ImageDataset
493         from torch.utils.data import DataLoader
494
495         # NB: this is WASB's OWN local `trackers` module (from its repo, on sys.path inside
496         the
497         # `wasb` conda env), NOT the Roboflow `trackers` package. The `type: ignore`
498         suppresses
499         # the mypy false positive where the main env resolves `trackers` to the Roboflow
500         package
501         # (which has no `build_tracker`). There is NO runtime collision ? this runs in a
502         separate
503         # subprocess/conda env. Do NOT "clean up" the ignore without removing the Roboflow
504         dep.
505         from trackers import build_tracker # type: ignore[attr-defined]
506         from utils import Center
507
508         from detectors import build_detector
509
510         streaming = frame_loader is not None
511
512         cfg = _build_cfg(weights, sport, device)
513         frames_in = int(cfg["model"]["frames_in"])
514         input_wh = (int(cfg["model"]["inp_width"]), int(cfg["model"]["inp_height"]))
515         output_wh = (int(cfg["model"]["out_width"]), int(cfg["model"]["out_height"]))
516
517         if streaming:
518             # `frames` is already the ordered list of (synthetic) frame paths.
519             if not isinstance(frames, (list, tuple)):
520                 raise SystemExit(
521                     "streaming mode requires `frames` to be a list of frame paths"
522                 )
523             frames = list(frames)
524             frames_dir = source_key or "stream"
525         else:
526             frames_dir = frames
527             frames = sorted(
528                 glob.glob(osp.join(frames_dir, "*.png"))

```



```

520         + glob.glob(osp.join(frames_dir, "*.jpg"))
521     )
522     if limit:
523         frames = frames[:limit]
524     if len(frames) < frames_in:
525         where = frames_dir if not streaming else "video stream"
526         raise SystemExit(f"need >= {frames_in} frames, found {len(frames)} in {where}")
527
528     # Sliding windows of `frames_in` consecutive frames (the unit of GPU work +
checkpoint).
529     samples = []
530     for window in build_windows(frames, frames_in):
531         annos = [
532             {"frame_path": p, "center": Center(is_visible=False, x=-1.0, y=-1.0)}
533             for p in window
534         ]
535         samples.append({"frames": window, "annos": annos})
536     windows_total = len(samples)
537
538     # ---- cache setup / resume decision ----- #
539     if cache_dir is None:
540         cache_dir = default_cache_dir(out_csv)
541     os.makedirs(cache_dir, exist_ok=True)
542     det_jsonl, _ = _cache_paths(cache_dir)
543     cache_key = source_key if source_key is not None else osp.abspath(frames_dir)
544     manifest = None if fresh else read_manifest(cache_dir)
545     resume = manifest_compatible(
546         manifest or {},
547         frames_dir=cache_key,
548         weights=weights,
549         sport=sport,
550         frames_in=frames_in,
551         frames_total=len(frames),
552         device=device,
553     )
554     if resume:
555         windows_done, det_by_fid = load_and_clean_cache(cache_dir)
556         logger.info(
557             f"resuming from cache: {windows_done}/{windows_total} windows already
detected"
558         )
559     else:
560         if manifest is not None:
561             logger.info("cache incompatible with this run -> starting fresh")
562             reset_cache(cache_dir)
563             windows_done, det_by_fid = 0, defaultdict(list)
564
565     base_manifest = {
566         "version": CACHE_VERSION,
567         "frames_dir": cache_key,
568         "weights": weights,
569         "sport": sport,
570         "frames_in": frames_in,
571         "frames_total": len(frames),
572         "device": device,
573         "windows_total": windows_total,
574         "windows_done": windows_done,
575     }
576     write_manifest(cache_dir, base_manifest)
577
578     # ---- detector pass over the un-cached windows ----- #
579     remaining = samples[windows_done:]
580     if remaining:
581         _, transform_test = build_img_transforms(cfg)
582         try:

```

```

583         _, seq_transform_test = build_seq_transforms(cfg)
584     except Exception:
585         seq_transform_test = None
586     ds = ImageDataset(
587         cfg,
588         remaining,
589         input_wh,
590         output_wh,
591         transform=transform_test,
592         seq_transform=seq_transform_test,
593         is_train=False,
594     )
595     # Streaming forces single-process loading: the in-memory frame store + the
596     # cv2.imread shim live in THIS process and can't be pickled to DataLoader
workers.
597     loader_workers = 0 if streaming else num_workers
598     loader = DataLoader(
599         ds, batch_size=batch_size, shuffle=False, num_workers=loader_workers
600     )
601     detector = build_detector(cfg)
602
603     from contextlib import ExitStack
604
605     t0 = time.time()
606     done = windows_done
607     win_idx = windows_done
608     log_every = max(1, log_every_batches)
609     flush_n = max(1, flush_every)
610     pending = []
611     logger.info(
612         f"detecting on {len(remaining)} remaining windows "
613         f"(total {windows_total}, batch={batch_size}, workers={loader_workers}, "
614         f"source={'video-stream' if streaming else 'frames-dir'})..."
615     )
616     det_f = open(det_jsonl, "a")
617     try:
618         with ExitStack() as stack:
619             stack.enter_context(torch.no_grad())
620             # In streaming mode, route ImageDataset's per-frame `cv2.imread(path)`
to the
621             # in-memory decoded frame for that synthetic path; every other line of
622             # __getitem__ runs unchanged so the tensor matches the PNG round-trip
exactly.
623             # The detector's first needed frame is `windows_done` (window i starts
at
624             # frame i), so prime the store there for a resume.
625             if streaming:
626                 stack.enter_context(
627                     _streaming_read_image_patch(
628                         frame_loader,
629                         len(frames),
630                         window=frames_in,
631                         start_index=windows_done,
632                     )
633                 )
634             # Overlap CPU batch assembly with GPU inference (streaming/workers=0
only ?
635             # a multi-worker DataLoader already overlaps via its worker processes).
636             batches = (
637                 _PrefetchIter(loader, depth=prefetch_batches)
638                 if prefetch_batches and loader_workers == 0
639                 else loader
640             )
641             for bi, batch in enumerate(batches):
642                 imgs, _hms, trans = batch[0], batch[1], batch[2]

```

```

643         img_paths = [
644             list(t) for t in batch[-1]
645         ] # [frame_in][batch] -> path
646         batch_results, _ = detector.run_tensor(imgs, trans)
647         nb = imgs.shape[0]
648         for ib in range(nb):
649             frames_payload = []
650             for ie in sorted(batch_results[ib].keys()):
651                 p = img_paths[ie][ib]
652                 fid = _frame_id(p)
653                 plain = [_det_plain(d) for d in batch_results[ib][ie]]
654                 frames_payload.append([fid, plain])
655                 det_by_fid[fid].extend(plain)
656             pending.append({"w": win_idx, "f": frames_payload})
657             win_idx += 1
658         done += nb
659         if len(pending) >= flush_n or done >= windows_total:
660             _append_windows(det_f, pending)
661             pending = []
662             base_manifest["windows_done"] = done
663             write_manifest(cache_dir, base_manifest)
664         if (bi + 1) % log_every == 0 or done >= windows_total:
665             el = time.time() - t0
666             new_done = done - windows_done
667             rate = new_done / el if el > 0 else 0.0
668             eta = (windows_total - done) / rate if rate > 0 else 0.0
669             logger.info(
670                 f"{done}/{windows_total} "
671                 f"({100 * done // max(1, windows_total)}%) | {rate:.1f}
win/s | "
672                 f"elapsed {el:.0f}s | ETA {eta:.0f}s"
673             )
674         if pending:
675             _append_windows(det_f, pending)
676             base_manifest["windows_done"] = done
677             write_manifest(cache_dir, base_manifest)
678         finally:
679             det_f.close()
680             logger.info(f"detection done in {time.time() - t0:.0f}s; running tracker...")
681     else:
682         logger.info("detector cache already complete -> tracker-only re-run")
683
684     # ---- tracker re-run over the full ordered cache (cheap, deterministic) ---- #
685     tracker = build_tracker(cfg)
686     tracker.refresh()
687     rows = []
688     for fid in sorted(det_by_fid.keys()):
689         r = tracker.update(_dets_to_tracker(det_by_fid[fid]))
690         rows.append((fid, 1 if r.get("visi") else 0, r.get("x", -1), r.get("y", -1)))
691
692     _atomic_write_trajectory(out_csv, rows)
693     visible = sum(1 for _, v, *_ in rows if v)
694     logger.info(f"wrote {len(rows)} rows -> {out_csv}")
695     logger.info(f"visible-shuttle frames: {visible}/{len(rows)}")
696     return out_csv
697
698
699 def extract_frames(video_path: str, frames_dir: str) -> str:
700     """Decode a video to PNG frames named by source frame index (00000000.png ...).
701
702     Idempotent: if the dir already holds the expected number of frames (matching the
703     container's reported frame count) we skip re-decoding ? re-extraction after a
704     reboot is wasteful. NOTE: cv2 PNG extraction is slow; for big clips prefer
705     extracting with ffmpeg on the host and passing --frames_dir.
706     """

```

```

707     import cv2
708
709     os.makedirs(frames_dir, exist_ok=True)
710     cap = cv2.VideoCapture(video_path)
711     if not cap.isOpened():
712         raise SystemExit(f"cannot open video: {video_path}")
713     expected = int(cap.get(cv2.CAP_PROP_FRAME_COUNT) or 0)
714     existing = len(glob.glob(osp.join(frames_dir, "*.png")))
715     if expected > 0 and existing >= expected:
716         cap.release()
717         logger.info(f"reusing {existing} already-extracted frames -> {frames_dir}")
718         return frames_dir
719     i = 0
720     while True:
721         ok, frame = cap.read()
722         if not ok:
723             break
724         cv2.imwrite(osp.join(frames_dir, f"{i:08d}.png"), frame)
725         i += 1
726     cap.release()
727     logger.info(f"extracted {i} frames -> {frames_dir}")
728     return frames_dir
729
730
731 def _probe_frame_count(video_path: str) -> int:
732     """Container-reported frame count via cv2 (0 if unknown). Used to size the
733     stream."""
734     import cv2
735
736     cap = cv2.VideoCapture(video_path)
737     if not cap.isOpened():
738         raise SystemExit(f"cannot open video: {video_path}")
739     n = int(cap.get(cv2.CAP_PROP_FRAME_COUNT) or 0)
740     cap.release()
741     return n
742
743 def make_video_frame_loader(video_path: str):
744     """Return ``(frame_loader, count_hint, release)`` for output-preserving streaming.
745
746     ``frame_loader(index)`` -> BGR uint8 ndarray`` decodes the video forward-only with one
747     long-lived ``cv2.VideoCapture``. It MUST be called with non-decreasing indices (the
748     detector's sliding-window order); it reads sequentially and only uses a frame-peek
749     to
750     skip forward when a *resume* starts past the current position. Each ``cap.read()``
751     returns exactly the BGR uint8 array that ``cv2.imwrite``/``cv2.imread`` of a PNG
752     would
753     yield (PNG is lossless), so the detector sees identical pixels to the disk path.
754
755     Returns the count hint from the container so the caller can build the frame list;
756     ``release`` closes the capture.
757     """
758     import cv2
759
760     cap = cv2.VideoCapture(video_path)
761     if not cap.isOpened():
762         raise SystemExit(f"cannot open video: {video_path}")
763     count_hint = int(cap.get(cv2.CAP_PROP_FRAME_COUNT) or 0)
764     state = {"pos": 0} # next index cap.read() will return
765
766     def frame_loader(index: int):
767         index = int(index)
768         if index < state["pos"]:
769             raise RuntimeError(
770                 f"streaming decode is forward-only; got index {index} after

```

```

{state['pos']}]}"
769         )
770         if index > state["pos"]:
771             # Forward skip only happens when resuming past cached windows. Seek once.
772             cap.set(cv2.CAP_PROP_POS_FRAMES, index)
773             state["pos"] = index
774             ok, frame = cap.read()
775             if not ok:
776                 raise RuntimeError(f"failed to decode frame {index} from {video_path}")
777             state["pos"] += 1
778             return frame
779
780     def release():
781         cap.release()
782
783     return frame_loader, count_hint, release
784
785
786 def run_video_streaming(
787     video_path: str,
788     weights: str,
789     out_csv: str,
790     sport: str = "badminton",
791     limit: int = 0,
792     batch_size: int = 8,
793     log_every_batches: int = 50,
794     cache_dir: "str | None" = None,
795     flush_every: int = 1000,
796     fresh: bool = False,
797     device: str = "cuda",
798     prefetch_batches: int = 0,
799 ) -> str:
800     """Fast path: decode ``video_path`` on the fly and feed frames straight to the
801     detector, WITHOUT extracting every frame to a PNG first (the disk round-trip that
802     dominates a long clip). Output is bit-identical to the frames-on-disk path; see
803     ``run``'s docstring for why.
804     """
805     frame_loader, count_hint, release = make_video_frame_loader(video_path)
806     try:
807         if count_hint <= 0:
808             raise SystemExit(
809                 f"could not determine frame count for {video_path}; cannot stream "
810                 f"(fall back to frame extraction with --frames_dir)"
811             )
812         frame_list = [synthetic_frame_path(i) for i in range(count_hint)]
813         source_key = "video:" + osp.abspath(video_path)
814         return run(
815             frame_list,
816             weights,
817             out_csv,
818             sport,
819             limit,
820             batch_size=batch_size,
821             num_workers=0,
822             log_every_batches=log_every_batches,
823             cache_dir=cache_dir,
824             flush_every=flush_every,
825             fresh=fresh,
826             frame_loader=frame_loader,
827             source_key=source_key,
828             device=device,
829             prefetch_batches=prefetch_batches,
830         )
831     finally:
832         release()

```

```

833
834
835 def main():
836     ap = argparse.ArgumentParser()
837     ap.add_argument("--frames_dir", help="folder of pre-extracted frames")
838     ap.add_argument(
839         "--video", help="video file; frames are extracted internally (in WSL)"
840     )
841     ap.add_argument(
842         "--frames_out_dir", help="where to extract frames when --video is used"
843     )
844     ap.add_argument(
845         "--stream-video",
846         action="store_true",
847         help="FAST PATH: decode --video on the fly and feed frames straight to "
848         "the detector (no PNG extraction). Output-equivalent to the disk path.",
849     )
850     ap.add_argument("--weights", required=True)
851     ap.add_argument("--out", required=True)
852     ap.add_argument("--sport", default="badminton")
853     ap.add_argument(
854         "--device",
855         default="cuda",
856         choices=["cuda", "mps", "cpu"],
857         help="torch device for the detector (default cuda = the WSL parity path)",
858     )
859     ap.add_argument(
860         "--limit", type=int, default=0, help="cap #frames (for quick tests)"
861     )
862     ap.add_argument("--batch-size", type=int, default=8)
863     ap.add_argument("--num-workers", type=int, default=4)
864     ap.add_argument(
865         "--prefetch-batches",
866         type=int,
867         default=0,
868         help="streaming mode: overlap CPU batch assembly with GPU inference via a "
869         "bounded producer thread of this queue depth (0 = off, the parity default)",
870     )
871     ap.add_argument(
872         "--log-every-batches", type=int, default=50, help="progress log cadence"
873     )
874     ap.add_argument(
875         "--cache-dir",
876         default=None,
877         help="resumable detector cache dir (default: <out>_wasbcache next to --out)",
878     )
879     ap.add_argument(
880         "--flush-every",
881         type=int,
882         default=1000,
883         help="fsync the detector cache every N windows (bounds reboot loss)",
884     )
885     ap.add_argument(
886         "--fresh", action="store_true", help="ignore any existing cache and start over"
887     )
888     args = ap.parse_args()
889
890     if args.stream_video:
891         if not args.video:
892             raise SystemExit("--stream-video requires --video")
893         run_video_streaming(
894             args.video,
895             args.weights,
896             args.out,
897             args.sport,

```

```

898         args.limit,
899         batch_size=args.batch_size,
900         log_every_batches=args.log_every_batches,
901         cache_dir=args.cache_dir,
902         flush_every=args.flush_every,
903         fresh=args.fresh,
904         device=args.device,
905         prefetch_batches=args.prefetch_batches,
906     )
907     return
908
909     frames_dir = args.frames_dir
910     if args.video:
911         frames_dir = extract_frames(
912             args.video, args.frames_out_dir or (args.video + "_frames")
913         )
914     if not frames_dir:
915         raise SystemExit("provide --frames_dir or --video")
916     run(
917         frames_dir,
918         args.weights,
919         args.out,
920         args.sport,
921         args.limit,
922         batch_size=args.batch_size,
923         num_workers=args.num_workers,
924         log_every_batches=args.log_every_batches,
925         cache_dir=args.cache_dir,
926         flush_every=args.flush_every,
927         fresh=args.fresh,
928         device=args.device,
929     )
930
931
932 if __name__ == "__main__":
933     main()
934

```

5. assistant (2026-07-03T19:48:31.973Z)

I have the two key files. Now let me read the config files to understand the existing knobs and how the argv is built, so I can reason about where TF32/cuDNN flags would slot in.

6. user (2026-07-03T19:48:34.907Z)

```

1     """Pydantic models for the application configuration.
2
3     This module defines the canonical, validated, typed configuration for the
4     Badminton Highlight Indexer. Built-in defaults live here (single source of truth).
5
6     Unknown / typo'd config keys are rejected at load time via extra="forbid".
7     """
8
9     from __future__ import annotations
10
11     from typing import Any, List, Literal, Optional
12
13     from pydantic import BaseModel, Field, field_validator, model_validator
14
15     Sport = Literal["badminton", "tennis", "pickleball", "table_tennis", "cricket"]
16
17
18     class ValidationRelevanceConfig(BaseModel):
19         court_green_lower: list[int] = Field(default=[35, 40, 40])
20         court_green_upper: list[int] = Field(default=[85, 255, 255])

```

```

21         court_pixels_min_ratio: float = Field(default=0.05, ge=0.0, le=1.0)
22
23
24     class ValidationConfig(BaseModel):
25         min_resolution_width: int = 1280
26         min_resolution_height: int = 720
27         min_fps: float = 29.9
28         min_blur_threshold: float = 20.0
29         max_scene_cuts_per_minute: float = 0.5
30         pts_max_gap_seconds: float = Field(
31             default=10.0,
32             ge=0.0,
33             description=(
34                 "Maximum allowed keyframe gap (seconds) before flagging as spliced. "
35                 "GoPro H.264/H.265 recordings typically have GOP intervals of 1?4s; "
36                 "a gap >10s suggests edited/missing content. Mobile phones, screen "
37                 "recordings, and other encoders may use different GOP sizes ? tune "
38                 "this threshold for your recording device."
39             ),
40         )
41         relevance: ValidationRelevanceConfig = Field(
42             default_factory=ValidationRelevanceConfig
43         )
44
45
46     class TrackNetConfig(BaseModel):
47         wsl_distro: str = "Ubuntu"
48         repo_dir: str = "~/models/TrackNetV4"
49         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
50         conda_env: str = "TrackNetV4"
51         weights_path: str = ""
52         queue_length: int = 5
53         stage_in_wsl: bool = True
54         wsl_stage_dir: str = "~/clips"
55         timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
56         velocity_threshold: float = 0.02
57         # Cache hygiene: after a SUCCESSFUL run the staged video copy (and, for WASB, the
58         # decoded frame cache) are deleted automatically ? they are pure intermediates,
59         # regenerable from the source, and the dominant disk consumer (~GBs/video). Set
60         # true only for debugging. On FAILURE they are always kept so a re-run can resume.
61         keep_frames: bool = False
62         # Absolute FLOOR (GB) on WSL free space (at wsl_stage_dir) before a run, on top of
63         # ~15x-source headroom the frame-stage guard always enforces (D11 ? it now BLOCKS,
64         # 0 = no extra floor (the headroom guard still applies).
65         min_free_gb: float = 0.0
66
67
68     class WasbIndexConfig(BaseModel):
69         """Typed config for the live WASB shuttle substrate (indexing.wasb).
70
71         Mirrors backend/pipeline/detectors/wasb_runner.py::WasbConfig.from_indexing_cfg so
72         these knobs actually take effect ? previously this whole block was silently dropped
73         because nested config sections defaulted to extra='ignore'.
74         """
75
76         wsl_distro: str = "Ubuntu"
77         repo_dir: str = "~/models/WASB-SBDT"
78         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
79         conda_env: str = "wasb"
80         weights_path: str = (
81             "~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar"
82         )
83         sport: str = "badminton"

```



```

84     wsl_stage_dir: str = "~/clips"
85     timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
86     velocity_threshold: float = 0.02
87     # --- Linux-native runner only (detector_impl='native'); ignored by the WSL runner.
88     # Env vars override these (WASB_DEVICE / WASB_PYTHON) for a config-free GPU box.
89     # torch device for the native runner. "auto" = cuda-if-available-else-cpu (the M4
CPU fallback
90     # for a no-GPU box); "cuda" stays strict (hard-fails without a GPU ? no silent slow-
CPU downgrade).
91     device: str = "cuda" # auto | cuda | mps | cpu
92     python_bin: str = (
93         "python" # the `wasb` conda env python (invoked by path, no `conda activate`)
94     )
95     # Cache hygiene: after a SUCCESSFUL run the staged video copy + decoded frame cache
96     # (the ~GBs/video disk hog) are deleted automatically ? pure intermediates,
regenerable
97     # from the source. Set true only for debugging. On FAILURE they are kept so a re-run
resumes.
98     keep_frames: bool = False
99     # Absolute FLOOR (GB) on WSL free space (at wsl_stage_dir) before a run, on top of
the
100    # ~15x-source headroom the frame-stage guard always enforces (D11 ? it now BLOCKS,
not warns).
101    # 0 = no extra floor (the headroom guard still applies).
102    min_free_gb: float = 0.0
103    # FAST PATH ? decode the video on the fly and feed frames straight to the detector,
104    # skipping the slow per-frame PNG extraction that dominates a long clip (~46k PNGs
for a
105    # 12-min 1080p60 video, which overran the 30-min timeout). Output is bit-identical
to the
106    # PNG path (PNG is lossless) ? VERIFIED on a real GPU 2026-06-20 (native stream ==
WSL disk,
107    # 1194/1195 frames byte-identical; native run-to-run byte-identical /
deterministic).
108    # Tri-state:
109    #     None          = per-runner default ? the WSL runner keeps DISK extraction
(unchanged Windows
110    #                     behaviour); the Linux-native runner (GPU box) defaults to STREAMING
(the perf win).
111    #     True/False = force on/off for BOTH runners (e.g. set False for a container cv2
can't seek).
112    stream_video: Optional[bool] = None
113    # GPU-feed tuning for the native runner (measured on the first real L4 serving run,
114    # 2026-07-03: the default batch 8 with single-process loading left the GPU <1%
utilized ?
115    # the pipeline was CPU-feed-bound). None = wasb_infer.py's parity defaults (batch 8,
no
116    # prefetch ? byte-identical to the historical path). The cloud-serving preset opts
into
117    # the measured fast values; local/WSL behaviour is unchanged unless set explicitly.
118    batch_size: Optional[int] = None
119    # Depth of the bounded producer-thread queue that overlaps CPU-side batch assembly
120    # (decode + PIL + ToTensor) with GPU inference in streaming mode. None/0 = off.
121    prefetch_batches: Optional[int] = None
122
123
124    class FusionConfig(BaseModel):
125        """Config for the multi-signal `fusion_hybrid` segmenter (MULTI_SIGNAL_FUSION_PLAN
P2).
126
127        Every default reproduces control behaviour: the fusion segmenter persists the
128        shuttle net-crossing count (Gate A signal) and ? only when ``cue_flags['person']``
is
129        true ? the person-cue features, but it does NOT gate the served handoff unless a

```

```

130     threshold is raised (`min_crossings`/`min_players` default 0 = OFF). The court
mask
131     is optional: `court_polygon=None` ? Gate B counts every detection (player-count-
only);
132     supplied ? off-court persons (spectators / adjacent courts) are excluded.
133     ""
134
135     # Which trajectory substrate seeds the windows (shuttle stays primary).
136     substrate: Literal["wasb", "tracknet"] = "wasb"
137     # Windowing velocity threshold for the fusion family (the engine reads
138     # indexing.<family>.velocity_threshold). Default matches the substrates' 0.02; set
139     # equal to indexing.wasb.velocity_threshold to A/B fusion against a tuned wasb run.
140     velocity_threshold: float = 0.02
141     # Per-cue enable flags, e.g. {"person": true}. Person cue is OFF unless explicitly
142     # enabled ? so the default path never imports torch / touches the GPU.
143     cue_flags: dict[str, bool] = Field(default_factory=dict)
144     # Served Gate A: drop windows whose shuttle net-crossings < this from the AI
handoff.
145     # 0 = OFF (no gating; persisted candidates are always the full superset for offline
re-scoring).
146     min_crossings: int = Field(default=0, ge=0)
147     # Served Gate B: when the person cue is on, drop windows with median in-court
players
148     # < this. 0 = OFF.
149     min_players: int = Field(default=0, ge=0)
150     # Optional court mask as normalised 0-1 [[x, y], ...] polygon. None = no mask.
151     court_polygon: Optional[list[list[float]]] = None
152     # Net line for the person two-sided-motion split (person features only ? the shuttle
153     # net-crossing gate keeps rally_gate's camera-agnostic auto-estimate).
154     net_axis: Literal["x", "y"] = "y"
155     net_position: float = Field(default=0.5, ge=0.0, le=1.0)
156
157
158     class IndexingConfig(BaseModel):
159         default_segmenter: str = "yolo_hybrid"
160         use_gpu: bool = True
161         # Detector backend for the trajectory hybrids (wasb/tracknet). "auto" = the real
162         # WSL runner (default, production); "stub" = a deterministic CPU mock (no GPU/WSL)
163         # so the whole pipeline is regression-testable on a CPU box / CI; "native" = the
164         # Linux-native WASB runner (no WSL/conda-activate; a GPU box ? M3.5). The
165         # RALLY_DETECTOR_IMPL env var overrides this. See
pipeline/detectors/{stub,native_wasb}_runner.py.
166         detector_impl: str = "auto"
167         motion_threshold: float = 0.08
168         min_segment_duration: float = 3.0
169         dead_time_merge_gap: float = 2.0
170         chunk_duration: float = 90.0
171         chunk_overlap: float = 10.0
172         detector_model: str = "fasterrcnn_resnet50_fpn"
173         detector_score_threshold: float = 0.5
174         yolo_velocity_threshold: float = 0.15
175         yolo_frame_skip: int = 3
176         yolo_tracker: str = "bytetrack.yaml"
177         yolo_prefetch_workers: int = 2
178         min_action_window_duration: float = 2.0
179         # BOUNDED WINDOWING (over-merge fix W1, RALLY_QUALITY_RESEARCH.md §6). 0 = OFF; > 0
=a window
180         # longer than this many seconds is recursively split at its largest internal
inactivity gap (?
181         # `window_min_split_gap` s ? the most likely rally boundary), so one window can't
swallow
182         # several rallies + the dead time between them. Never merges, only cuts (recall
can't drop).
183         # *** R1 MILESTONE DEFAULT-ON (cap=6): the smallest safe local-windowing flip.
Measured n=13

```

```

184      # golden, R2 tolF1: baseline?milestone (cap=6 + R4 below) ?+0.222, sign 12/0,
p=0.0005; cap=6
185      # beats cap=12 ?+0.110, sign 12/0; net over-seg guard PASS (mean merge_rate
0.651?0.161). ***
186      max_window_duration: float = 6.0
187      window_min_split_gap: float = (
188          0.5 # min internal gap (s) that qualifies as a split point
189      )
190      # HARD-CAP fallback for W1: when True, an over-long window with NO internal
inactivity gap
191      # (a continuously-active trajectory ? e.g. the real-footage mahadevpura-1 174s blob)
is
192      # chopped at its midpoint until ? max_window_duration, making the cap a true upper
bound.
193      # Only cuts (recall can't drop). OFF by default until measured/promoted.
194      window_hard_cap: bool = False
195      # R4 rally-END inactivity trim (s, 0 = OFF). After a rally the shuttle keeps moving
slowly
196      # (picked up / walked) above velocity_thresh, so the window over-extends its END
(the measured
197      # |?end| gap vs golden). Trim the post-rally tail back to the last frame whose
trailing
198      # window stays dense with FAST motion (velocity > inactivity_rally_thresh). Only
cuts.
199      # *** R1 MILESTONE DEFAULT-ON (1.5/0.20/0.30): on top of cap=6, R4 adds a small but
significant
200      # end-precision gain ? ?tolF1 +0.040, sign 9/1/3, p=0.022, |?end| 4.96?3.31s (n=13).
The W1 cap
201      # is the dominant lever; R4 is the marginal increment. See QUALITY_ITERATIONS.md
"R1". ***
202      window_inactivity_end: float = 1.5
203      inactivity_rally_thresh: float = 0.20
204      inactivity_min_density: float = 0.30
205      # R4 density DENOMINATOR fix (code-review finding B). The R4 trim's fast-fraction
denominator is
206      # the COUNT of active samples in the trailing window, which sparse tracking inflates
(one fast
207      # sample after a gap ? 100% dense ? tail held open). When True, the denominator is
the window's
208      # TEMPORAL length instead. No-op under dense tracking (sample-count == temporal-
span), so it only
209      # corrects the sparse-track case. *** P2b PROMOTED DEFAULT-ON (2026-06-29): golden-
set A/B at the
210      # SERVED operating point (n=15, 595 rallies) ? tolF1 0.353?0.364 (?tolF1 +0.010,
sign 10W/4L, NO
211      # regression), |?end| 2.97?2.72s, seg_count_ratio ?0.103 (p=0.006), split_rate
?0.010 (p=0.016),
212      # R1 over-seg guard PASS; served_regresses(eps=0.0)=False ?
promote_if_served_safe(structural)=True.
213      # See docs/RALLY_DETECTION_FIXES_PLAN.md §7/§8. ***
214      inactivity_temporal_density: bool = True
215      # R5 blob-fallback (default OFF). LAST-RESORT splitter for the continuously-active
blob: after
216      # the W1 gap-split AND the R4 inactivity trim have each had their chance, a group
STILL longer
217      # than window_blob_factor × max_window_duration (no internal gap, no rally-end
signal ? e.g.
218      # mahadevpura-1's ~65s residual ? 11× a 6s cap) is midpoint-chopped to ? the cap as
a FORCED cut
219      # (logged + flagged on the window) so the degenerate single-window outcome is
avoided and the
220      # clip is surfaced as needing rally-STATE cues, not a bigger cap. Differs from
window_hard_cap
221      # (which chops BEFORE R4 and over-segments normal clips): blob-fallback runs AFTER
R4 and only

```

```

222     # force-cuts the genuine blob (the factor gate). Only cuts (recall can't drop).
223     window_blob_fallback: bool = False
224     # Magnitude gate that keeps R5 surgical: only a group longer than this ×
max_window_duration is a
225     # blob (a normal long rally ~1-2× the cap is left alone; the mahadevpura-1 residual
is ~11×).
226     # Default 4.0 is do-no-harm on the golden corpus (rescues only the continuously-
active
227     # clips, every other clip a no-op within noise) ? see docs/QUALITY_ITERATIONS.md
"R5".
228     window_blob_factor: float = 4.0
229     log_yolo_candidates: bool = True
230     store_yolo_candidates: bool = True
231     ai_handoff_padding: float = 2.0
232     ai_max_workers: int = 5
233     # Width (px) the gemini segmenter re-encodes chunks to before upload. Stream-copied
234     # chunks of a high-bitrate (4K) source blow past the provider upload cap
235     # (backend/providers/gemini.py::MAX_UPLOAD_MB) and are refused ? observed live as
236     # "0 segments / success". Re-encoding also makes chunk boundaries frame-accurate
237     # (stream copy cuts at the nearest keyframe). 0 = legacy stream-copy at source res.
238     # Matches gemini_refine's --clip-scale-width default.
239     ai_chunk_scale_width: int = 1280
240     # FULLY-LOCAL mode (default OFF): when true, the trajectory-hybrid segmenters
241     # (wasb_hybrid / tracknet_hybrid / fusion_hybrid) skip the Phase-3 AI handoff
entirely and
242     # emit their local candidate windows AS the rallies ? no provider, no API key,
nothing
243     # uploaded. Lets the local detector run standalone (e.g. to benchmark it against the
Gemini
244     # path, or to operate with zero cloud dependency). The pure `gemini` segmenter
ignores this.
245     skip_ai_handoff: bool = False
246     # Optional debug knob: trim every video to the first N seconds before processing.
247     debug_duration: Optional[float] = None
248
249     tracknet: TrackNetConfig = Field(default_factory=TrackNetConfig)
250     wasb: WasbIndexConfig = Field(default_factory=WasbIndexConfig)
251     fusion: FusionConfig = Field(default_factory=FusionConfig)
252
253     @field_validator("default_segmenter")
254     @classmethod
255     def segmenter_must_be_registered(cls, v: str) -> str:
256         # Lazy import: the segmenter registry imports modules that may import
257         # config, so we resolve it at validation time rather than at module load.
258         from backend.pipeline.segmenters.base import segmenter_registry
259
260         if segmenter_registry.get(v) is None:
261             available = sorted(segmenter_registry.list_available().keys())
262             raise ValueError(
263                 f"Segmenter '{v}' is not registered. Available: {'', '.join(available)}"
264             )
265         return v
266
267     @model_validator(mode="after")
268     def _chunk_step_must_be_positive(self) -> "IndexingConfig":
269         # The chunked segmenters advance by (chunk_duration - chunk_overlap); a non-
positive
270         # step makes `while t < duration: t += step` loop forever, growing memory before
any
271         # GPU work. Reject the bad config at load time instead.
272         if self.chunk_overlap >= self.chunk_duration:
273             raise ValueError(
274                 f"indexing.chunk_overlap ({self.chunk_overlap}) must be < chunk_duration
"
275                 f"({self.chunk_duration}); otherwise chunking never advances."

```

```

276         )
277         return self
278
279
280 class OutputConfig(BaseModel):
281     default_highlights_name: str = "highlights.mp4"
282     db_name: str = "sports_indexer.db"
283     # Directory where the DB, highlight reels, and processed clips are written.
284     # The CLI's --output-dir overrides this at startup (see backend/main.py:main).
285     output_dir: str = "output"
286
287
288 class WatermarkConfig(BaseModel):
289     """Branding overlay burned into each highlight clip during the per-clip re-encode
290     (backend/pipeline/stitcher/compiler.py). **On by default** ? set `enabled: false`
291     (under `stitching.watermark` in config.json) to turn it off. The text is fully
292     configurable. The concat stage stays stream-copy (-c copy)."""
293
294     enabled: bool = True
295     text: str = "Generated by Khelsutra.guru"
296     logo_path: str = "" # optional transparent PNG overlay; "" = no logo
297     font_path: str = "" # optional .ttf; "" = use the fontconfig `font` family below
298     font: str = "Sans" # fontconfig family used when font_path is empty
299     font_size_ratio: float = Field(
300         default=0.035, gt=0.0, le=0.5
301     ) # text height as a fraction of frame height
302     opacity: float = Field(default=0.85, ge=0.0, le=1.0)
303     box: bool = True # faint backing box behind the text for legibility
304     margin: int = Field(default=24, ge=0) # px inset from the chosen corner
305     position: Literal["bottom-right", "bottom-left", "top-right", "top-left"] = (
306         "bottom-right"
307     )
308
309
310 class StitchingConfig(BaseModel):
311     begin_padding: float = 0.5
312     end_padding: float = 0.5
313     use_cache: bool = False
314     min_shot_count: int = 1
315     # Long-rally reel filter (seconds; 0 = OFF). Drop detected rally pairs whose
duration
316     # (end ? start) is below this from the highlight reel, BEFORE compiling. A cleaner
317     # reel-selection knob than `min_shot_count`: shot_count is unlabeled / "Unknown" on
the
318     # fully-local no-AI path, whereas duration is derived from the rally's own
start/end. The
319     # local detector over-fires and its false positives are mostly SHORT (motion blips,
320     # background-court fragments), so this keeps the reel to the eye-catching long
rallies.
321     # OFF by default ? the detector output is unchanged, only which rallies reach the
reel.
322     min_rally_duration: float = Field(default=0.0, ge=0.0)
323     watermark: WatermarkConfig = Field(default_factory=WatermarkConfig)
324
325
326 class LoggingConfig(BaseModel):
327     """Logging knobs for the run entrypoints (see backend/utils/logging_setup.py)."""
328
329     # Third-party HTTP/RPC client loggers (httpx, httpcore, urllib3, google-genai,
330     # google.auth, grpc) emit one INFO line per request ? dozens per Gemini run, which
331     # buries our own [rally]/[compile] lines in run.log during manual QA. Default raises
332     # them to WARNING; set true to restore full chatter for request-flow debugging.
333     verbose_http: bool = False
334
335

```

```

336     class SecurityConfig(BaseModel):
337         # When set, /api/process video_path must resolve under this directory (hosted/tenant
338         # mode). Default None preserves the single-user local 'any local file' workflow.
339         allowed_video_root: Optional[str] = None
340
341         # --- Chunked upload (B2, PR-2) ---
342         # Landing zone for browser uploads (assembled from parts). Per-user subdirectories
343         # arrive with PR-3 identity scoping. ? In hosted mode, set allowed_video_root to
344         # contain upload_dir or /api/process will reject the uploaded paths.
345         upload_dir: str = "output/uploads"
346         # Whole-file cap (default 4 GB) and per-part cap. Parts stay under ~100 MB because
347         # free-tier edge proxies cap request bodies there (HOSTING_PLAN $2).
348         max_upload_mb: int = 4096
349         max_part_mb: int = 95
350         allowed_upload_extensions: List[str] = ["mp4", "mov"]
351
352         # --- M9 edge auth (Cloudflare Access, D9) ---
353         # DEFAULT-OFF: edge_auth_enabled=False ? no JWT is required/verified (the local
single-user
354         # path is unchanged). When True, /api/process requires + verifies the `Cf-Access-
Jwt-Assertion`
355         # header against the team JWKS (so a direct hit to the Cloud Run URL can't spoof the
identity
356         # header) and tags the job with the verified user_id (owner_id). The team domain
(e.g.
357         # ``https://<team>.cloudflareaccess.com``) supplies the JWKS + issuer; the AUD is
the CF Access
358         # application's audience tag. NO secrets here ? these are public identifiers.
359         edge_auth_enabled: bool = False
360         cf_team_domain: str = (
361             "" # e.g. https://<team>.cloudflareaccess.com (issuer + /certs JWKS)
362         )
363         cf_access_aud: str = "" # the CF Access application AUD tag the token must carry
364         # (CORS allow-origins is set via the RALLY_CORS_ALLOW_ORIGINS env var ? read at
import in
365         # backend/main.py so importing the app does no filesystem I/O; default [""].
Credentials stay
366         # OFF ? Cf-Access is a header, not a cookie.)
367
368
369     class StorageConfig(BaseModel):
370         """Pluggable storage backend (docs/DECOUPLED_COMPUTE/). Default ``local`` = today's
371         behaviour (filesystem paths, byte-for-byte). ``s3`` covers AWS + Cloudflare R2 + D0
Spaces
372         + MinIO (via ``endpoint_url``); ``gcs`` = Google Cloud Storage; ``gdrive`` = a
locally MOUNTED
373         Google Drive (``gdrive.mount_root``). Per-backend settings are loose dicts passed to
the constructor ?
374         **secrets are never stored here**, only endpoints / regions / file paths / source-
selectors
375         (boto3/google auth chains supply credentials). See ``backend/storage/``. """
376
377         backend: Literal["local", "s3", "gcs", "gdrive", "gdrive-api"] = "local"
378         # Output root for produced artifacts. "" => fall back to output.output_dir (non-
breaking).
379         output_root: str = ""
380         # --- Input side (COMPUTE_DECOUPLED_SERVING M2; parallel to the output
`backend`/`output_root`).
381         # Default ``local`` / "" keeps today's behaviour: a bare path / ``local://`` URI is
read IN PLACE
382         # (identity, no copy). Set these to read inputs from a bucket / mounted Drive ? a
bare/relative
383         # input name is resolved against ``input_root`` (e.g.
``input_root``="gs://bucket/videos"`` + "x.mp4"
384         # => ``gs://bucket/videos/x.mp4``) or, if ``input_root`` is empty, prefixed with

```

```

`input_backend`
385     # (`"gcs"` + "bucket/x.mp4" => `gcs://bucket/x.mp4`). An explicit scheme on the
input
386     # (`gs://?`/`s3://?`) or an absolute local path ALWAYS wins. Non-local inputs
are fetched to a
387     # scratch dir before processing (see `backend.storage.acquire_local_input`).
388     input_backend: Literal["local", "s3", "gcs", "gdrive", "gdrive-api"] = "local"
389     input_root: str = ""
390     # --- Per-video workspace (docs/PER_VIDEO_WORKSPACE.md) ---
391     # One folder per video keyed by the MD5 video_id:
<root>/[<workspace_prefix>]/<video_id>/...
392     # so a local deploy and a cloud (bucket) deploy share ONE relative layout (only the
root
393     # differs). DEFAULT-OFF: convergence is phased + flipped via a Launch Report (D5).
When OFF
394     # the legacy flat layout is used unchanged.
395     single_folder_per_video: bool = False
396     # Durable workspace root URI. "" => precedence: output_root, then output.output_dir.
397     workspace_root: str = ""
398     # Cloud namespace prefix under the root so per-video folders sit tidily beside
399     # videos/ labels/ weights/ (e.g. gs://bucket/workspaces/<video_id>/). "" = root
directly.
400     workspace_prefix: str = "workspaces"
401     local: dict[str, Any] = Field(default_factory=dict)
402     s3: dict[str, Any] = Field(default_factory=dict)
403     gcs: dict[str, Any] = Field(default_factory=dict)
404     gdrive: dict[str, Any] = Field(default_factory=dict)
405     # gdrive-api settings (e.g. {"root_folder_id": "<id>"}); read via the
hyphen?underscore lookup in
406     # backend.storage.get_storage. NO secrets ? creds come from ADC over the Drive
scope.
407     gdrive_api: dict[str, Any] = Field(default_factory=dict)
408
409
410     class DeploymentConfig(BaseModel):
411         """Deployment profile ? one switch that selects a coherent bundle of compute +
storage
412         knobs (COMPUTE_DECOUPLED_SERVING M1, `docs/COMPUTE_DECOUPLED_SERVING/README.md`
§4.3).
413
414         **Pure / additive:** the empty default profile (`""`) applies NO preset, so the
pipeline
415         behaves exactly as before this section existed. A profile is opt-in (`config.json`
`deployment.profile` or the `DEPLOYMENT_PROFILE` env var); auto-detect only ever
416         *suggests* one (D15 ? cloud spend is never entered silently). Nothing in the core
reads
417         these fields yet (M1): selecting a profile works purely by pre-setting the EXISTING
knobs
418         (`indexing.detector_impl` / `skip_ai_handoff`, `storage.backend`) in the
loader.
419
420         Valid values mirror `backend/config/presets.py` (parity-pinned by a unit test)."""
421
422         # "" = no profile (today's manual knobs). Others apply presets.PROFILE_PRESETS.
423         profile: Literal["", "truly-local", "cloud-serving", "cpu-mock"] = ""
424         # Where the core runs. Set by the profile preset; overridable in config.json. Inert
in M1
425
426         # (recorded for observability; consumed by the ComputeBackend seam from M2+).
427         compute_target: Literal["", "local_gpu", "cloud_run", "cpu_mock"] = ""
428
429
430     class BillingConfig(BaseModel):
431         """Per-job cost ledger (COMPUTE_DECOUPLED_SERVING M8, D8).
432
433         **DEFAULT-OFF:** `enabled=False` ? nothing records cost (the v6 cost columns stay

```

```

NULL =
434         today's behaviour). When enabled, a job's cost is computed + stored in **USD** via
the versioned
435         ``pricebook`` DB table and **displayed in INR** at ``fx_inr_per_usd``. The default
pricebook
436         version is the seeded ``placeholder-v0`` (every rate ``0.0`` ? cost ``0``) until the
owner
437         inserts a real, calibrated version and points ``pricebook_version`` at it.
438
439         ? Even with ``enabled=True`` + a real pricebook, recorded cost stays **0** until a
follow-up
440         wires the worker to emit per-job ``usage`` (gpu/wall/egress/gemini) ? M8 lands the
ledger +
441         pricebook + the recording seam; usage emission (the metering hooks) is a separate
step.
442
443         enabled: bool = False
444         pricebook_version: str = (
445             "placeholder-v0" # the seeded $0 version; owner flips to a real one
446         )
447         fx_inr_per_usd: float = 83.0 # USD?INR display rate (Bangalore market)
448         margin: float = 0.0 # resale markup (?0); 0 = bill at cost
449
450
451     class FlywheelConfig(BaseModel):
452         """Data-flywheel harness (COMPUTE_DECOUPLED_SERVING M11, D12) ? the
consent?capture?eval?goldenize
453         loop: on a CONSENTED adult job, durably **capture** the source + rally output into
the per-video
454         workspace ? **shadow-eval** owned-vs-Gemini ? **propose promotion** to the golden
TRAINING set, so
455         the owned model climbs toward the D11 ship floor.
456
457         **DEFAULT-OFF + INERT:** ``enabled=False`` ? nothing is captured/evaluated (byte-
identical to
458         today). ? **Consent is FAKED-DENY until counsel + the consent schema land:** even
``enabled=True``
459         captures **nothing**, because ``flywheel.consent_attested`` currently returns
``False`` for every
460         job (the privacy guardrail ? D10: adults-only, explicit opt-in, derived-data + auto-
delete-row).
461         This is the PLUMBING only; the real consent gate (counsel-reviewed ToS/DPA ? consent
columns ?
462         wiring) + the owned-model + golden-label data needed to *activate* shadow-
eval/promotion are
463         owner/counsel/GPU scope. Pairs with M9-consent.
464
465         enabled: bool = False
466         workspace_prefix: str = (
467             "flywheel" # per-video workspace prefix for captured artifacts
468         )
469         shadow_eval: bool = True # run the owned-vs-Gemini shadow comparison on capture
470         min_promote_n: int = 5 # promote_if_served_safe `min_n` (the served-safe gate)
471
472
473     class AppConfig(BaseModel):
474         """Root configuration object.
475
476         All runtime configuration should be accessed via an instance of this class
477         (or the loader) rather than raw dict lookups.
478
479
480         sport: Sport = "badminton"
481         ai_provider: str = "gemini"
482         # Auto-tracking alias: 2.5-flash is legacy now that 3.5-flash is the stable

```



```

flagship;
483         # the alias never deprecates underneath us. Eval reports record the run date
alongside.
484         ai_model: str = "gemini-flash-latest"
485
486         validation: ValidationConfig = Field(default_factory=ValidationConfig)
487         indexing: IndexingConfig = Field(default_factory=IndexingConfig)
488         output: OutputConfig = Field(default_factory=OutputConfig)
489         stitching: StitchingConfig = Field(default_factory=StitchingConfig)
490         security: SecurityConfig = Field(default_factory=SecurityConfig)
491         logging: LoggingConfig = Field(default_factory=LoggingConfig)
492         storage: StorageConfig = Field(default_factory=StorageConfig)
493         deployment: DeploymentConfig = Field(default_factory=DeploymentConfig)
494         billing: BillingConfig = Field(default_factory=BillingConfig)
495         flywheel: FlywheelConfig = Field(default_factory=FlywheelConfig)
496
497         # Reject unknown keys ? typed config migration is complete (D3).
498         model_config = {
499             "extra": "forbid",
500             "validate_assignment": True,
501         }
502
503
504         # Convenience type alias for callers that want to annotate
505         Config = AppConfig
506

```

7. user (2026-07-03T19:48:35.563Z)

```

1         """Deployment-profile presets + environment auto-detection (COMPUTE_DECOUPLED_SERVING
M1).
2
3         A *deployment profile* is a single switch that applies a coherent BUNDLE of existing
4         config knobs, so the truly-local / cloud-serving / cpu-mock modes don't drift apart one
5         knob at a time (the design in ``docs/COMPUTE_DECOUPLED_SERVING/README.md`` §4.3).
6
7         Precedence (lowest ? highest), realised by the loader::
8
9             built-in model defaults < profile preset < config.json < env < worker --set
10
11         This module is **pure data + pure functions** ? no IO, and (critically) **no torch
import at
12         module load**. The loader applies a preset ONLY when a profile is EXPLICITLY selected
13         (``config.json`` ``deployment.profile`` or the ``DEPLOYMENT_PROFILE`` env var); with no
14         profile selected the loader is byte-identical to the pre-M1 behaviour (``profile=""`` ==
15         today). Auto-detect only ever **suggests** a profile ? it never auto-applies one (D15:
cloud
16         spend is never entered silently).
17         """
18
19         from __future__ import annotations
20
21         import copy
22         import os
23         from typing import Any, Mapping, Optional
24
25         # Canonical enum values, defined ONCE here. ``models.DeploymentConfig``'s ``Literal``
mirrors
26         # these and ``tests/test_config_deployment.py`` pins parity so the two can never drift.
27         PROFILES: tuple[str, ...] = ("truly-local", "cloud-serving", "cpu-mock")
28         COMPUTE_TARGETS: tuple[str, ...] = ("local_gpu", "cloud_run", "cpu_mock")
29
30         # Each profile maps 1:1 to its canonical compute_target. Used to apply the preset and to
warn
31         # when config.json explicitly overrides compute_target into a state inconsistent with

```

the profile.

```
32     COMPUTE_TARGET_FOR_PROFILE: dict[str, str] = {
33         "truly-local": "local_gpu",
34         "cloud-serving": "cloud_run",
35         "cpu-mock": "cpu_mock",
36     }
37
38     # A preset is a partial, nested override dict over the model defaults. It touches ONLY
knobs
39     # that EXIST today ? M1 is pure/additive: the new
``deployment.{profile,compute_target}`` plus
40     # ``indexing.{detector_impl,skip_ai_handoff}`` and ``storage.backend``. Later milestones
extend
41     # these bundles ? ``input_backend`` (M2) and the configurable output/workspace base (M3,
42     # Launch-Report-gated) ? deliberately NOT set here so M1 never pre-empts a gated flip.
43     PROFILE_PRESETS: dict[str, dict[str, Any]] = {
44         "truly-local": {
45             "deployment": {"profile": "truly-local", "compute_target": "local_gpu"},
46             # detector_impl stays "auto" (the runner picks WSL vs native by env on the GPU
box).
47             # Zero-cloud / privacy-first ? run fully local with no Gemini handoff.
48             "indexing": {"skip_ai_handoff": True},
49             "storage": {"backend": "local"},
50         },
51         "cloud-serving": {
52             "deployment": {"profile": "cloud-serving", "compute_target": "cloud_run"},
53             # Linux L4 (cuda); Gemini handoff ON until the owned model clears the #172 floor
(D11).
54             # wasb GPU-feed tuning: the parity defaults (batch 8, no prefetch) left the paid
L4
55             # <1% utilized on the first real serving run (2026-07-03) ? the pipeline was
56             # CPU-feed-bound. Serving opts into the measured fast values; local/WSL
unchanged.
57             "indexing": {
58                 "detector_impl": "native",
59                 "skip_ai_handoff": False,
60                 "wasb": {"batch_size": 64, "prefetch_batches": 4},
61             },
62             "storage": {"backend": "gcs"},
63         },
64         "cpu-mock": {
65             "deployment": {"profile": "cpu-mock", "compute_target": "cpu_mock"},
66             # Deterministic CPU mock ? no GPU/WSL, no provider/API. The CI / regression
profile.
67             "indexing": {"detector_impl": "stub", "skip_ai_handoff": True},
68             "storage": {"backend": "local"},
69         },
70     }
71
72
73     def is_known_profile(profile: str) -> bool:
74         """True for a non-empty, recognised profile name."""
75         return bool(profile) and profile in PROFILE_PRESETS
76
77
78     def preset_for(profile: str) -> dict[str, Any]:
79         """Return a deep COPY of the preset bundle for ``profile`` (so callers can't mutate
the
80         module-level table), or ``{}`` for ``""`` / an unknown profile."""
81         return copy.deepcopy(PROFILE_PRESETS.get(profile, {}))
82
83
84     def _truthy(val: Optional[str]) -> bool:
85         """Common env-var truthiness (``1/true/yes/on``, case/space-insensitive)."""
86         if val is None:
```

```

87         return False
88     return val.strip().lower() in {"1", "true", "yes", "on"}
89
90
91     def _cuda_available() -> bool:
92         """Best-effort CUDA probe. torch is heavy + optional, so it is imported lazily here
and
93         ONLY when a caller opts into ``probe_cuda``; any import/runtime failure means 'no
GPU'."""
94         try:
95             # Intentionally lazy + local: keeps the config-load path torch-free (torch is
heavy
96             # and optional). Only reached when a caller opts into probe_cuda.
97             import torch
98
99             return bool(torch.cuda.is_available())
100     except Exception:
101         return False
102
103
104     def probe_cuda_available() -> bool:
105         """Public best-effort CUDA probe (lazy torch import; ``False`` on any failure).
106
107         Exposed for the per-profile startup checks (``backend/config/startup.py``) so a
caller
108         that *wants* the GPU heads-up can pass the result in without the startup module
importing
109         torch itself. Best-effort by design: if torch lives only in a detector subprocess
env
110         (native WASB), this returns ``False`` in the main process ? which is why the startup
checks
111         treat a GPU mismatch as a WARNING (the runner healthcheck is the authority), never a
hard
112         error that could false-fail."""
113         return _cuda_available()
114
115
116     def suggest_profile(
117         env: Optional[Mapping[str, str]] = None, *, probe_cuda: bool = False
118     ) -> str:
119         """SUGGEST (never auto-apply ? D15) a deployment profile from the environment.
120
121         Signals, in priority order:
122
123         - ``K_SERVICE`` set (Cloud Run) ? ``"cloud-serving"``
124         - ``CI`` truthy ? ``"cpu-mock"``
125         - ``probe_cuda`` and a CUDA GPU present ? ``"truly-local"``
126         - otherwise ? ``""`` (no strong suggestion)
127
128         ``probe_cuda`` defaults to ``False`` so the common config-load path stays cheap and
never
129         imports torch; a CLI ``--detect`` or cloud-serving startup that wants the GPU probe
passes
130         ``probe_cuda=True``. The result is advisory only: the caller decides whether to act
on it.
131         """
132         e = os.environ if env is None else env
133         if e.get("K_SERVICE"):
134             return "cloud-serving"
135         if _truthy(e.get("CI")):
136             return "cpu-mock"
137         if probe_cuda and _cuda_available():
138             return "truly-local"
139         return ""
140

```

8. user (2026-07-03T19:48:36.436Z)

```
1      # Cloud Run GPU **Job** image for the rally worker (COMPUTE_DECOUPLED_SERVING M6/M7,
ADR-012/D16).
2      #
3      # Runs `python -m backend.worker infer ?` INLINE on an L4 (detect + window + stitch co-
located so
4      # CPU/wall + egress are metered into one job ? D4). Built + pushed + verified by the
owner on the
5      # M7 real run (the $100/?10k budget, D17); CI does NOT build this (a CUDA image is too
heavy) ? the
6      # dispatch shim is proven with a fake client (tests/test_compute_backend.py).
7      #
8      # Two python stacks, deliberately separate (a framework conflict otherwise ? same split
as the
9      # native GPU box, deploy/digitalocean/gpu_setup.sh):
10     # * the MAIN app env (this repo + ffmpeg) drives the pipeline + stitching;
11     # * a conda `wasb` env (py3.8 + torch 1.11.0+cu113) runs the WASB shuttle detector as
a subprocess
12     # (the native runner shells out to $WASB_PYTHON), so the app's torch never co-
installs with it.
13     #
14     # ? WASB-C3: the pretrained badminton weights are academic-data-trained ? provenance NOT
cleared for
15     # commercial sharing. They are NOT baked into this image; stage them at runtime (a
bucket fetch or a
16     # mount) and point $WASB_WEIGHTS at them. Resolve WASB-C3 before any public, non-owner
serving.
17
18     FROM nvidia/cuda:12.4.1-runtime-ubuntu22.04
19
20     ENV DEBIAN_FRONTEND=noninteractive \
21         PYTHONUNBUFFERED=1 \
22         PIP_NO_CACHE_DIR=1 \
23         CONDA_DIR=/opt/conda \
24         WASB_REPO_DIR=/opt/models/WASB-SBDT
25
26     # ffmpeg (stitch/decode) + DejaVu fonts (the watermark filter) + git/curl for the WASB
env build.
27     RUN apt-get update && apt-get install -y --no-install-recommends \
28         ffmpeg fonts-dejavu-core git curl ca-certificates build-essential \
29         python3 python3-pip python3-venv \
30         && rm -rf /var/lib/apt/lists/*
31
32     # --- WASB detector env (micromamba + conda-forge ? BSD-licensed, NO Anaconda ToS; P3a,
ADR-005) ---
33     # Enforces ADR-005 (every dependency ? code, data, weights, AND the build toolchain ?
stays
34     # MIT/Apache/BSD): micromamba (BSD-3) + conda-forge (community) replaces Miniconda's
Anaconda-ToS-gated
35     # `defaults` channels (the ?200-employee Terms-of-Service trap flagged in
PACKAGING_PLAN.md §3).
36     # MAMBA_ROOT_PREFIX=$CONDA_DIR (= /opt/conda) keeps the env at $CONDA_DIR/envs/wasb so
$WASB_PYTHON below
37     # is UNCHANGED. Same verified torch-1.11.0+cu113 WASB stack; the pip wheels are
unaffected by the switch.
38     ENV MAMBA_ROOT_PREFIX=/opt/conda
39     RUN curl -sSL https://micro.mamba.pm/api/micromamba/linux-64/latest \
40         | tar -xj -C /usr/local bin/micromamba \
41         && micromamba create -y -q -n wasb -c conda-forge python=3.8 \
42         && git clone --depth 1 https://github.com/nttcom/WASB-SBDT.git "$WASB_REPO_DIR" \
43         && "$CONDA_DIR/envs/wasb/bin/python" -m pip install -q \
44             torch==1.11.0+cu113 torchvision==0.12.0+cu113 --extra-index-url
https://download.pytorch.org/whl/cu113 \
45         && ( [ -f "$WASB_REPO_DIR/requirements.txt" ] && "$CONDA_DIR/envs/wasb/bin/python"
```

```

-m pip install -q -r "$WASB_REPO_DIR/requirements.txt" || true ) \
46      && "$CONDA_DIR/envs/wasb/bin/python" -m pip install -q \
47      hydra-core omegaconf pandas opencv-python-headless pillow scipy tqdm pyyaml
matplotlib "numpy<1.24"
48
49      # --- Main application env ---
50      WORKDIR /app
51      COPY pyproject.toml README.md ./
52      COPY requirements-trackers.txt ./
53      COPY backend ./backend
54      COPY training ./training
55      # Install with the [storage-gcs] extra: the cloud worker PULLS the gs:// input and
PUSHES the
56      # gs:// outputs/done-marker through backend.storage (google-cloud-storage is lazy-
imported there).
57      # Without it the job dies at the first gs:// fetch with `ModuleNotFoundError:
google.cloud` ? the
58      # real-Cloud-Run failure the mock-client tests can't see (the bucket I/O is faked in
tests).
59      # D13/P12: `trackers` (ByteTrack's new home) is a git-only dep (broken PyPI wheel ?
60      # roboflow/trackers#474) so it's NOT pulled by `pip install .`; install it out-of-band
before
61      # the main env, or the person-cue path crashes with ModuleNotFoundError. The immutable
git SHA is
62      # single-sourced in requirements-trackers.txt.
63      RUN python3 -m pip install --upgrade pip \
64      && python3 -m pip install -r requirements-trackers.txt \
65      && python3 -m pip install ".[storage-gcs]"
66
67      # The native runner finds the WASB env + repo by env (no `conda activate` needed);
weights come in at
68      # runtime (WASB-C3). detector_impl=native = the Linux-native WASB runner; cuda stays the
default.
69      # PYTHONUNBUFFERED: without it the worker's stdout/stderr sit in the block buffer until
process
70      # exit, so Cloud Logging shows a single end-of-run burst instead of live progress
(observed on the
71      # first real M7 runs, 2026-07-03) ? a running job looks stalled and mid-run failures log
nothing.
72      ENV PYTHONUNBUFFERED=1 \
73      WASB_PYTHON=/opt/conda/envs/wasb/bin/python \
74      WASB_REPO_DIR=/opt/models/WASB-SBDT \
75      WASB_WEIGHTS=/opt/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar \
76      RALLY_DETECTOR_IMPL=native
77
78      # The dispatcher (CloudRunCompute) supplies the per-execution args:
79      # infer --video-uri gs:/// --out-uri gs:/// --done-uri gs:/// --segmenter wasb_hybrid
--set ?
80      ENTRYPOINT ["python3", "-m", "backend.worker"]
81

```

9. assistant (2026-07-03T19:48:41.963Z)

I now have the full picture. Let me verify the critical constraint about the torch version in the WASB env, because TF32 behavior is highly version-dependent ? torch 1.11.0 predates several TF32-related default changes. This is load-bearing for the analysis.

The Dockerfile confirms: WASB env is **torch 1.11.0+cu113, Python 3.8**. This is critical because:

- In torch 1.11, `torch.backends.cuda.matmul.allow\_tf32` defaults to `False`, but `torch.backends.cudnn.allow\_tf32` defaults to `True` (the historical asymmetric default that existed 1.7?1.11).
- `torch.backends.cudnn.benchmark` defaults to `False`.

Let me confirm there's no existing setting of these flags anywhere in the infer script or repo,

and check what device the parity was validated on.

10. user (2026-07-03T19:48:45.713Z)

<persisted-output>

Output too large (79.5KB). Full output saved to: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\tool-results\toolu_01Pbv1VH63p2kNYKyKXjcLz4.txt

Preview (first 2KB):

```
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\README.md:362:Once a video is processed, you can measure **how well the pipeline detected its rallies** against hand-annotated ground truth. You annotate each rally's `[start, end)` time range in a small CSV; the harness scores the detected rallies straight from the SQLite DB ? no API calls, fully deterministic.
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\ABLATION_LAYER.md:76:The rest of the suite runs GPU-free on deterministic synthetic candidate rows.
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\ANALYZERS_AND_RECONCILERS.md:93:binary verdict ? the small-N-safe way to fold in a deterministic-ish cue.
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\ANALYZERS_AND_RECONCILERS.md:103: deterministic, zero production risk.
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\ANALYZERS_AND_RECONCILERS.md:133:class BoundaryHint: #
OPTIONAL deterministic proposal ? never auto-applied
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\ANALYZERS_AND_RECONCILERS.md:372:occupancy, analyzer signals/events, session spans, shuttle-state) ? it is a deterministic linter, not a
...
</persisted-output>
```

11. assistant (2026-07-03T19:48:48.515Z)

The output is dominated by "deterministic" doc mentions. Let me narrow to just the actual torch backend flag calls in Python code, which is what matters for whether these flags are already set.

12. user (2026-07-03T19:48:50.730Z)

No matches found

13. user (2026-07-03T19:48:52.583Z)

Found 6 files

```
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\pipeline\detectors\wasb_infer.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\pipeline\detectors\native_wasb_runner.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\pipeline\detectors\stub_runner.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\pipeline\detectors\tracknet_runner.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
```

```
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-  
worktree\backend\pipeline\detectors\wasb_runner.py  
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-  
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-  
worktree\backend\pipeline\detectors\person_detector.py
```

14. assistant (2026-07-03T19:49:03.187Z)

No existing calls to any of these three flags anywhere in the Python code ? confirming they are all currently at their library defaults inside the WASB subprocess. This is the key fact for the "safe-on / opt-in / unsafe" verdict.

Let me confirm the parity guarantee's exact wording and the key detail: the native runner's own module docstring already states cuDNN is non-deterministic ACROSS GPUs but byte-exact run-to-run on a FIXED GPU. That distinction is central to the benchmark analysis.

I have everything I need. Let me record the key confirmed facts before writing the finding:

1. `**torch 1.11.0+cu113**` in the WASB subprocess env (Dockerfile lines 44). This version's TF32 defaults: ``matmul.allow_tf32=False`, `cudnn.allow_tf32=True`` (the historical asymmetric default), ``cudnn.benchmark=False``.
2. `**No flag is set anywhere**` in ``wasb_infer.py`` or the runner ? all three are at library defaults inside the subprocess.
3. The forward runs under ``torch.no_grad()`` in ``wasb_infer.py::run`` (line 619), calling ``detector.run_tensor(imgs, trans)`` ? a conv-heavy hourglass/HRNet backbone, so conv is the dominant op (cudnn territory), with some matmul.
4. The parity gate (native_wasb_runner.py docstring, lines 26-34): parity is asserted `**within ~1e-3 px float tolerance across GPUs**`, but `**byte-exact run-to-run on a FIXED GPU**`, and `**two native runs were byte-identical (deterministic)**`. This is the load-bearing property.
5. The prompt's own D5/D16 constraint: output must stay `**bit-identical/deterministic across runs and across disk-frames vs streaming paths**`.

Now I'll write the structured finding.

15. user (2026-07-03T19:49:41.264Z)

```
<tool_use_error>InputValidationError: StructuredOutput was called with input that could not be  
parsed as JSON.  
You sent (first 200 of 6479 bytes): {"lever": "tf32-cudnn: three independent torch backend  
flags, set once at wasb_infer.py process start (before build_detector). Verdict differs per flag  
? cudnn.benchmark=True is the only real win and  
Common causes: unescaped backslashes in file paths (use / or \\), unescaped control characters,  
or truncated output. Retry with valid JSON.</tool_use_error>
```

16. assistant (2026-07-03T19:49:44.961Z)

The tool expects the fields as top-level structured arguments, not a raw JSON string. Let me call it correctly.

17. user (2026-07-03T19:50:15.229Z)

Output does not match required schema: root: must have required property 'mechanism', root: must have required property 'expected_win', root: must have required property 'parity_risk', root: must have required property 'code_locations', root: must have required property 'implementation_sketch', root: must have required property 'confidence'

18. assistant (2026-07-03T19:50:18.625Z)

The parameters after ``parity_risk`` were not parsed ? the ``code_locations`` value contains a ``">`` sequence that broke the argument boundary. Let me remove the stray quote/angle-bracket characters from that field and resubmit cleanly.

19. user (2026-07-03T19:50:42.819Z)

Output does not match required schema: root: must have required property 'mechanism', root: must have required property 'expected_win', root: must have required property 'parity_risk', root: must have required property 'code_locations', root: must have required property 'implementation_sketch', root: must have required property 'confidence'

20. assistant (2026-07-03T19:50:47.595Z)

The schema still reports those fields as missing even though I passed them. This strongly suggests the harness is concatenating the XML-style parameters and one field's content is breaking parsing ? likely a stray characters issue. Let me try shorter, plainer values with no problematic punctuation (no `::`, no parentheses-heavy nesting that might confuse the parser), keeping each field concise.

21. user (2026-07-03T19:51:05.422Z)

Output does not match required schema: root: must have required property 'mechanism', root: must have required property 'expected_win', root: must have required property 'parity_risk', root: must have required property 'code_locations', root: must have required property 'implementation_sketch', root: must have required property 'confidence'

22. assistant (2026-07-03T19:51:13.401Z)

The parameters keep getting dropped. The pattern is that only `lever` (the first param) survives, and everything from `mechanism` onward is missing ? this looks like a parsing failure where the tool call isn't receiving the multiline parameters correctly. Let me try a single JSON object passed as one argument by making the parameter values very short first, to isolate whether it's a size/multiline issue.

23. user (2026-07-03T19:51:15.302Z)

<tool_use_error>InputValidationError: StructuredOutput was called with input that could not be parsed as JSON.
You sent (first 200 of 236 bytes): {"lever": "test", "mechanism": "test mechanism", "expected_win": "test win", "parity_risk": "determinism-changing", "gated_on": "none", "code_locations": "wasb_infer.py:run", "implementation_sketch": "t
Common causes: unescaped backslashes in file paths (use / or \\), unescaped control characters, or truncated output. Retry with valid JSON.</tool_use_error>